

DSC 411/511
Advance Data Mining Algorithms and Applications

Fall 2026

Instructor: Ijaz Ul Haq, Ph.D.

ijazhaq@umich.edu

Location: MSB 104



Lecture 05
Classification
(Naive Bayes, k-NN)

Previous class:

- ***What is classification?***
- ***Decision tress***
- ***Related NoteBook (code)***

Today's class:

- **Naive Bayes**
  - ***Where to use it?***
  - ***How it works?***
- **k-Nearest Neighbor (k-NN)**
  - ***Where to use it?***
  - ***How it works?***
- **Related NoteBook (code)**

Classification: Naïve Bayes

Definition:

A naive Bayes classifier estimates the probability of each class given the attribute values, assuming that the attributes are conditionally independent of one another given the class.

Chapter 4
Introduction to Data Mining, 2nd Edition

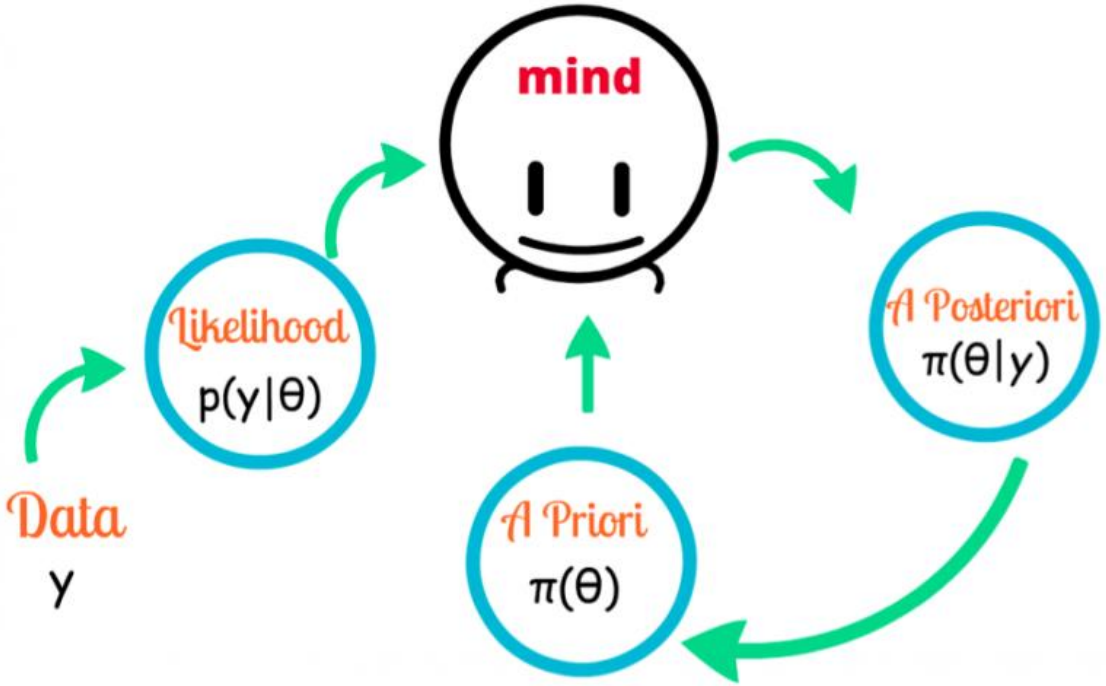


Image Source: https://blog.datumbox.com/machine-learning-tutorial-the-naive-bayes-text-classifier/

Classification: Naïve Bayes

Where it can be used?

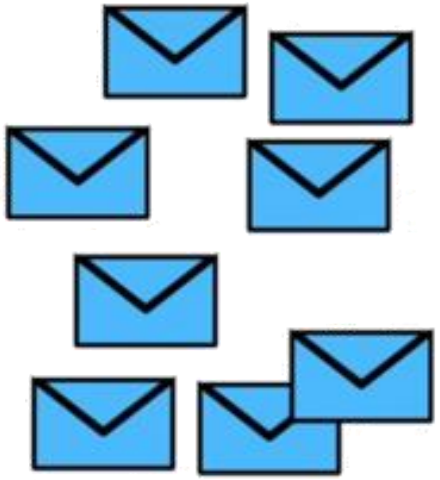
- **Spam filtering.** The original mass-market application. Still competitive.
- **Document and text classification.** Thousands of word columns, and it stays fast.
- **Real-time systems.** Training is one pass over the data. Prediction is a multiplication.

How it works?

- From the training data, count **how common each** class is.
- For each column, learn how its **values are distributed** inside each class.
- For a new row, compute the **probability of each class**.
- Multiply the per-column probabilities together.
- Pick the class with the highest result.

Classification: Naïve Bayes

The Working of Naïve Bayes



Imagine we received normal email everyday from friends and family.

Classification: Naïve Bayes

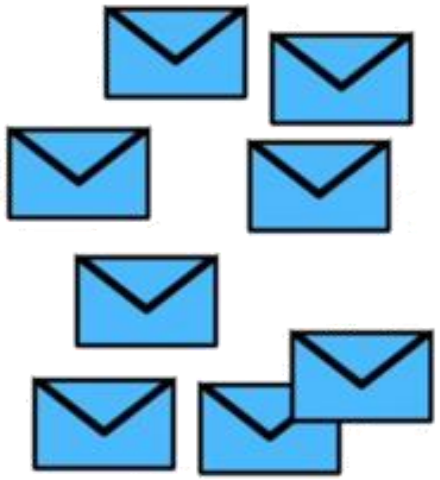
The Working of Naïve Bayes



… But we also receive spam (unwanted emails that are usually scams)…

Classification: Naïve Bayes

The Working of Naïve Bayes



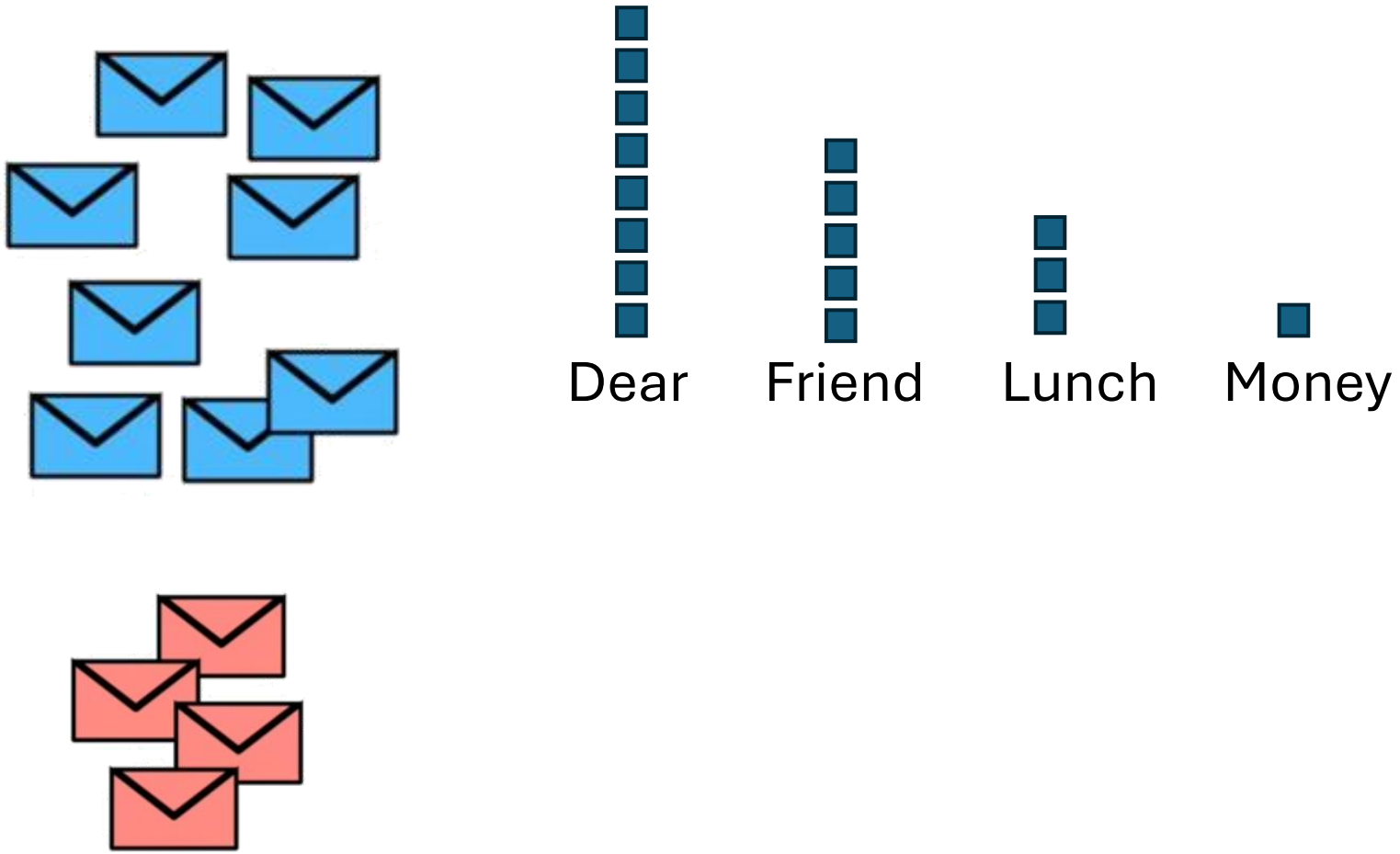


… So the task is to filter out the **spam** emails.

How can we do this?

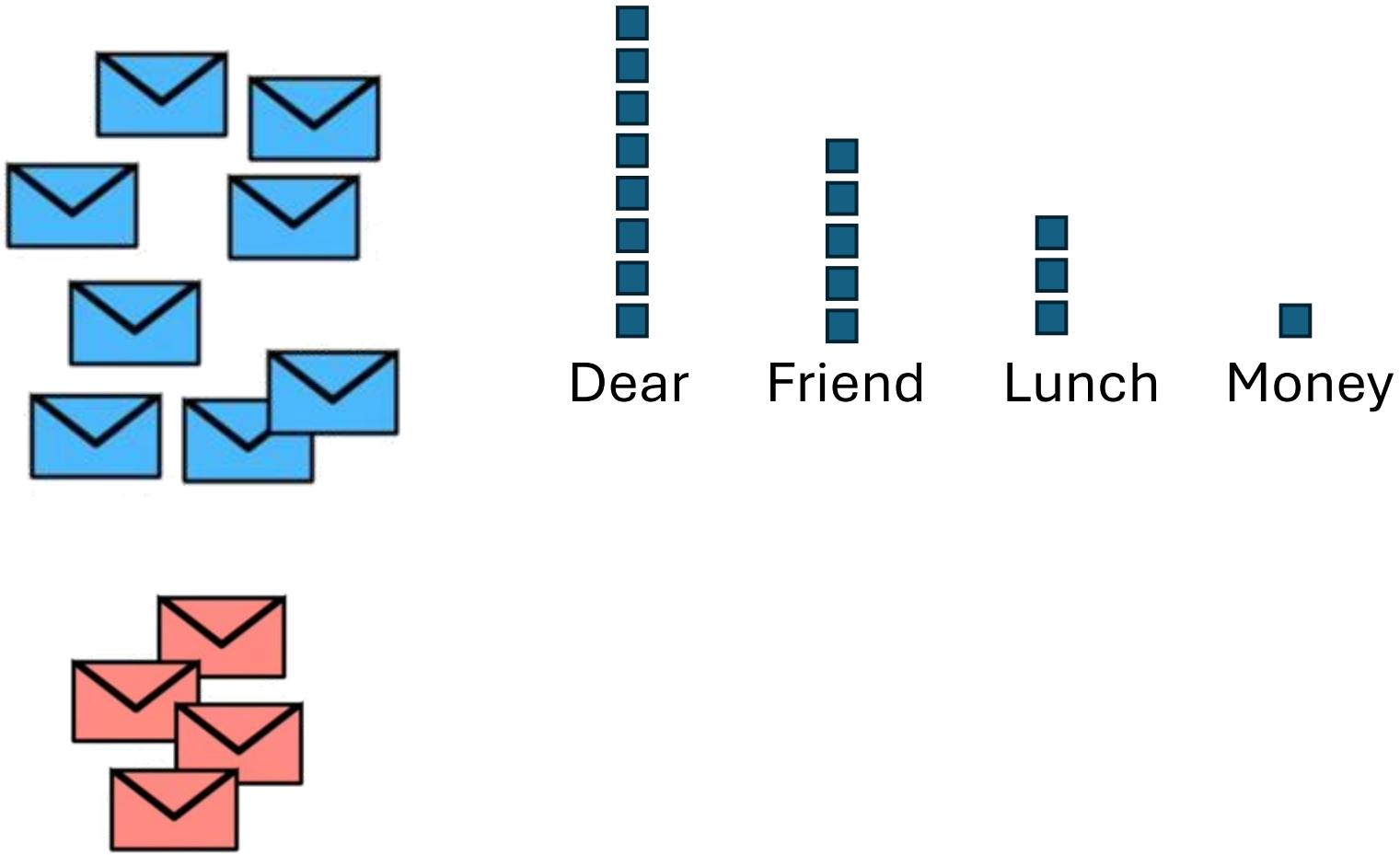
Classification: Naïve Bayes

The Working of Naïve Bayes



Classification: Naïve Bayes

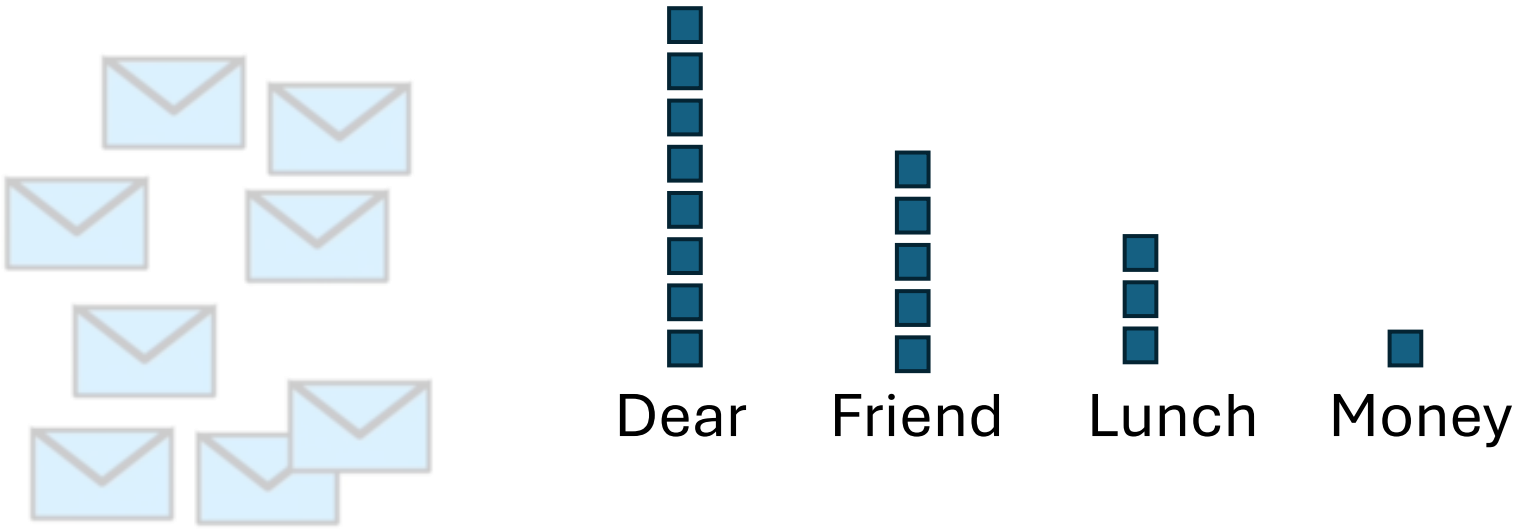
The Working of Naïve Bayes



Make a histogram of all the words that occur in the normal emails

Classification: Naïve Bayes

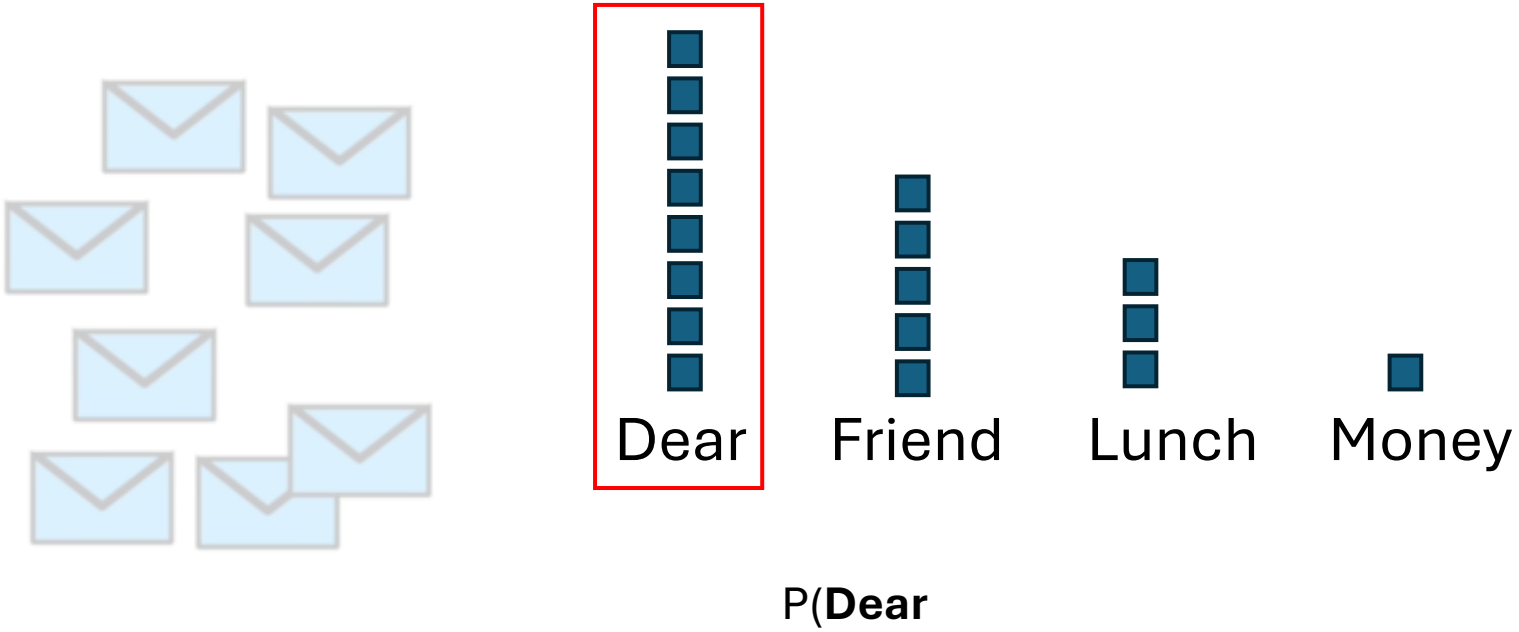
The Working of Naïve Bayes



Use histogram of each word to calculate the probabilities of seeing each word, given that it was in a normal message.

Classification: Naïve Bayes

The Working of Naïve Bayes

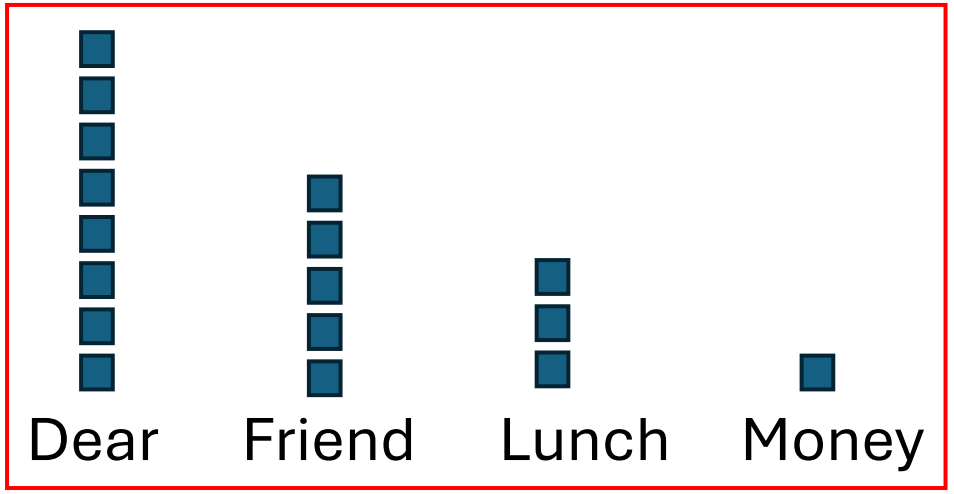


P(**Dear**

Use histogram of each word to calculate the probabilities of seeing each word, given that it was in a normal message.

Classification: Naïve Bayes

The Working of Naïve Bayes

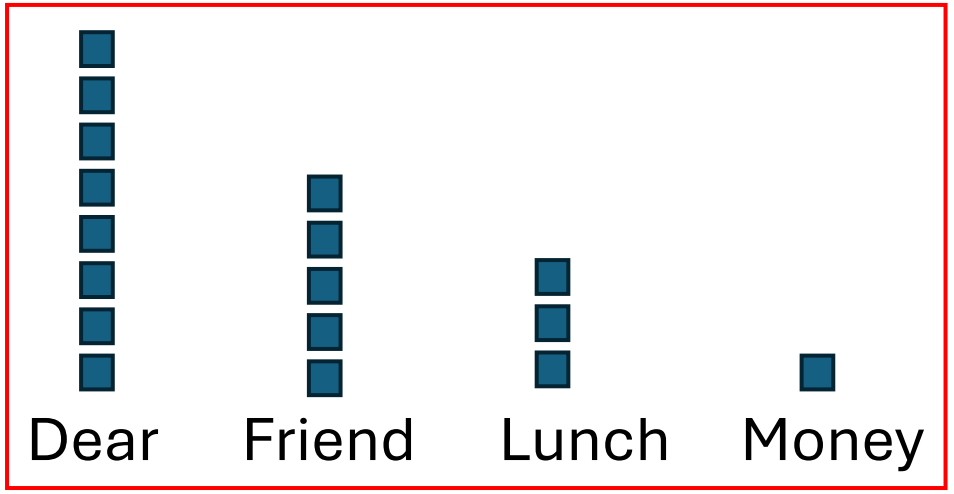


Use histogram of each word to calculate the probabilities of seeing each word, given that it was in a normal message.

P(**Dear** | **Normal**) = 8 /17

Classification: Naïve Bayes

The Working of Naïve Bayes

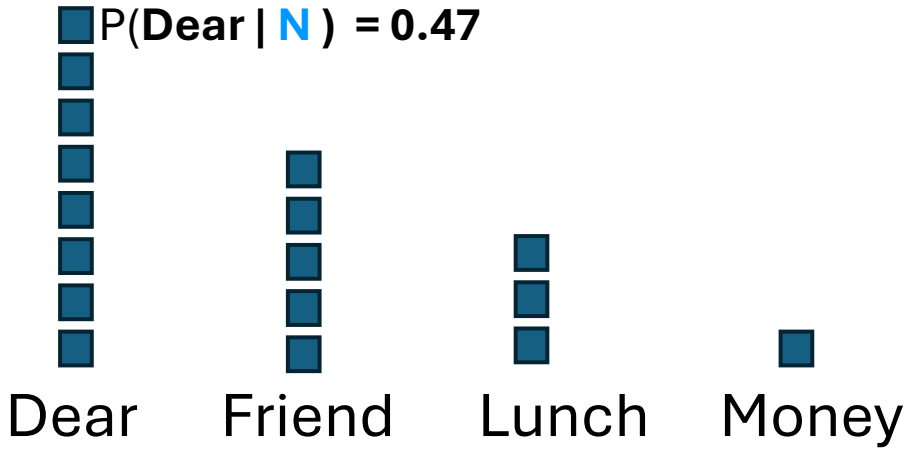


Use histogram of each word to calculate the probabilities of seeing each word, given that it was in a normal message.

P(**Dear** | **Normal**) = 8 /17 = 0.47

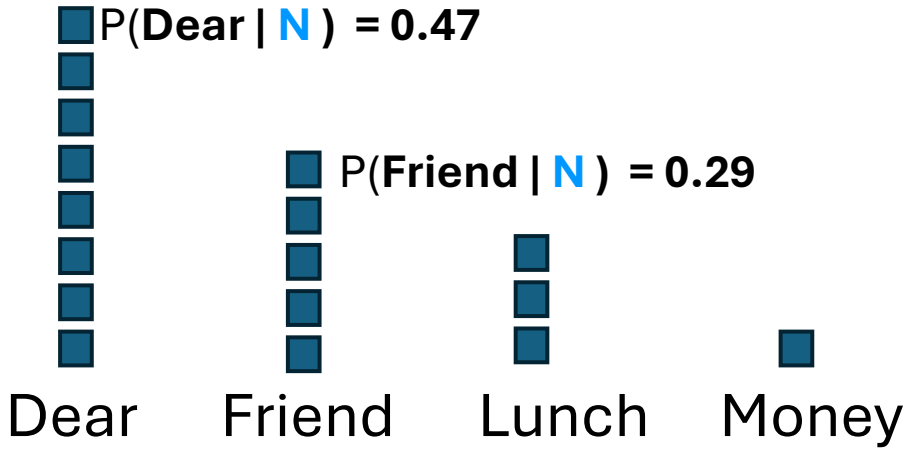
Classification: Naïve Bayes

The Working of Naïve Bayes



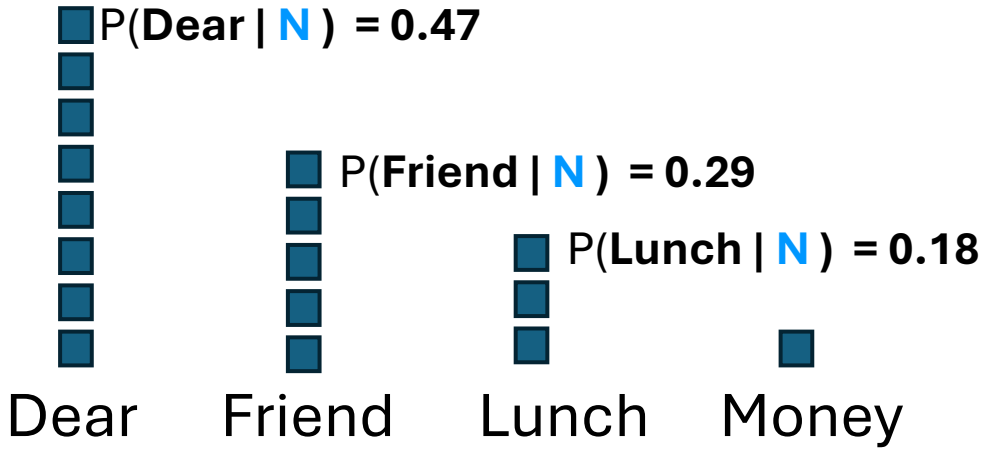
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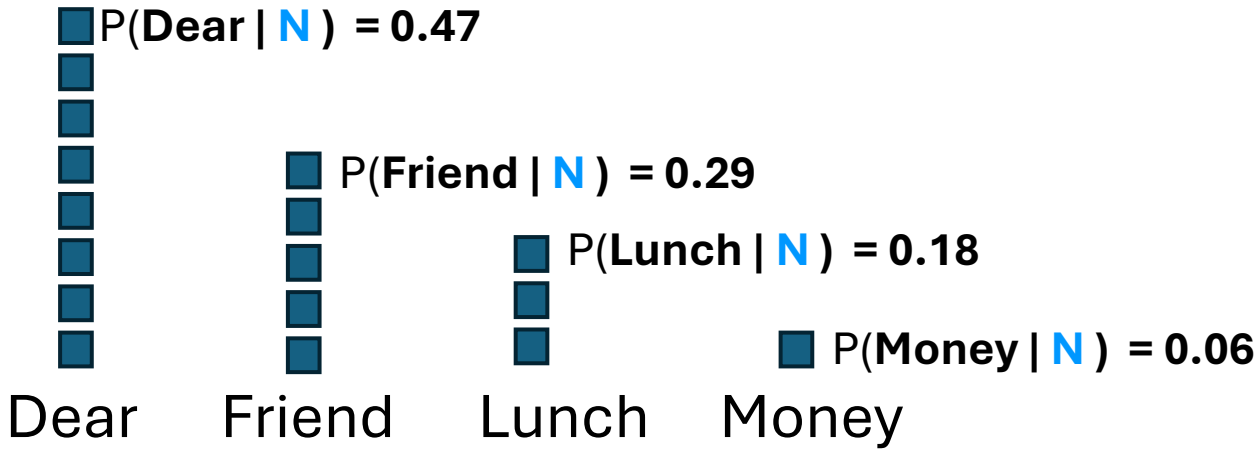
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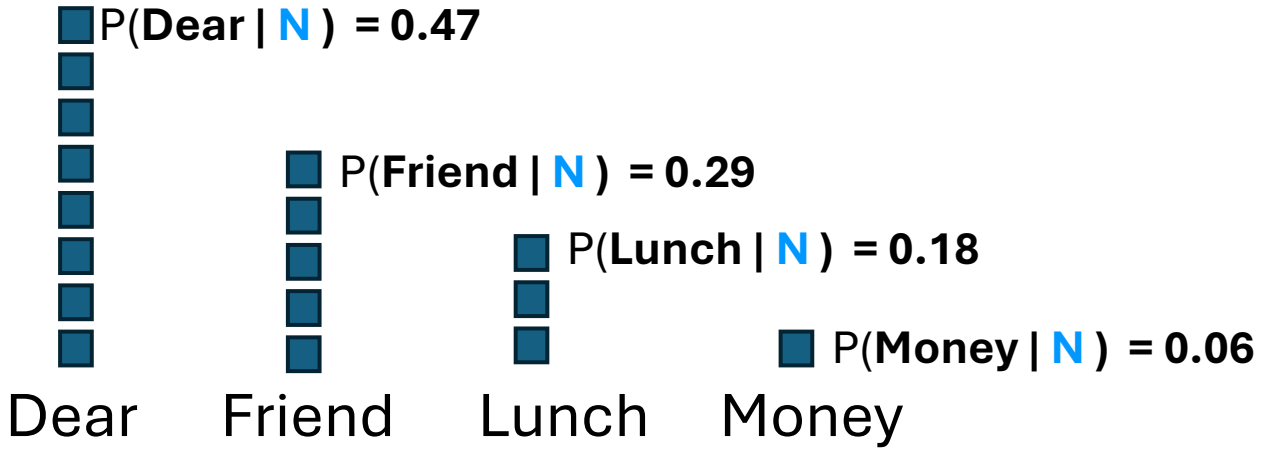
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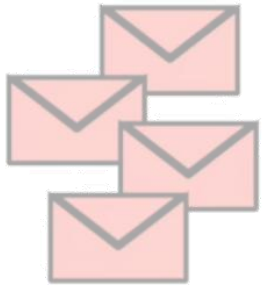
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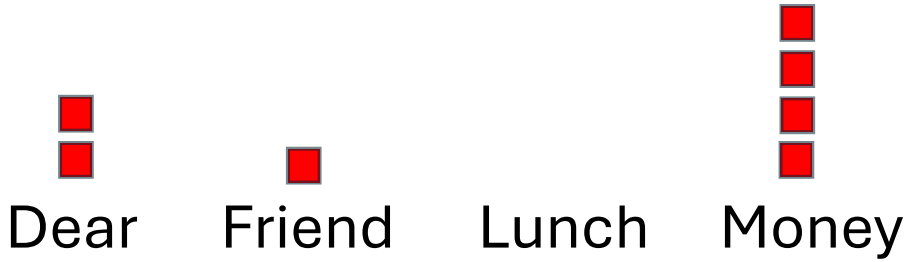


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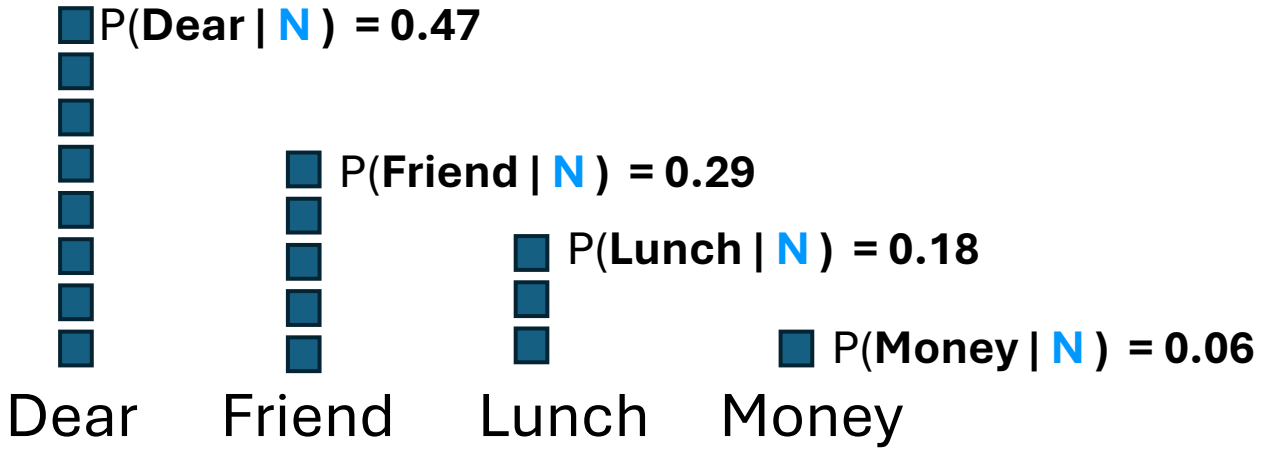


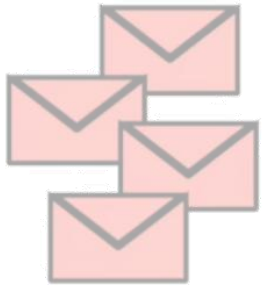


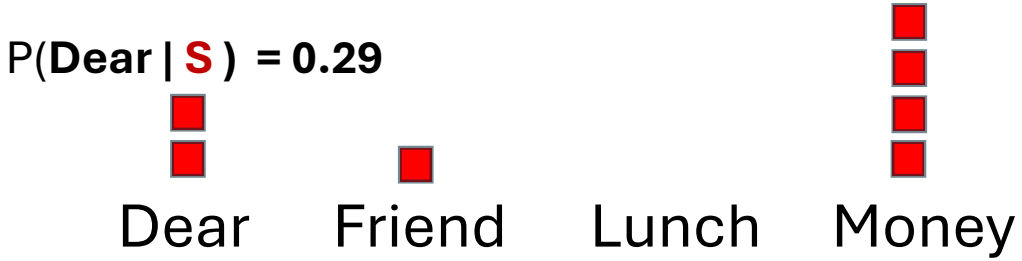


Classification: Naïve Bayes

The Working of Naïve Bayes

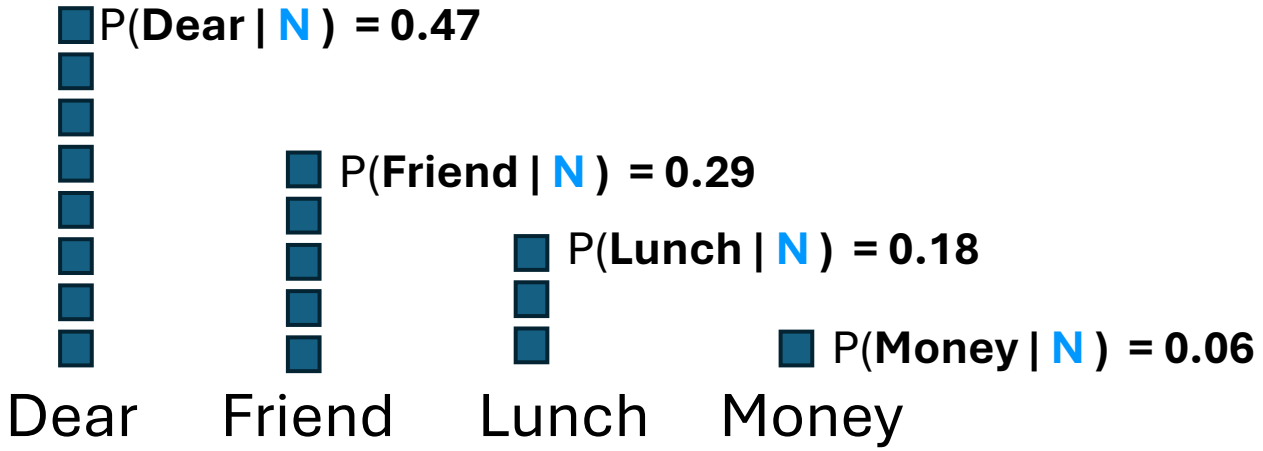


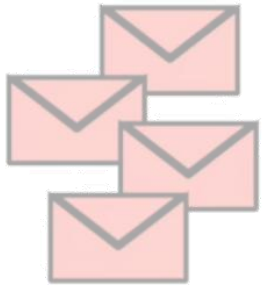


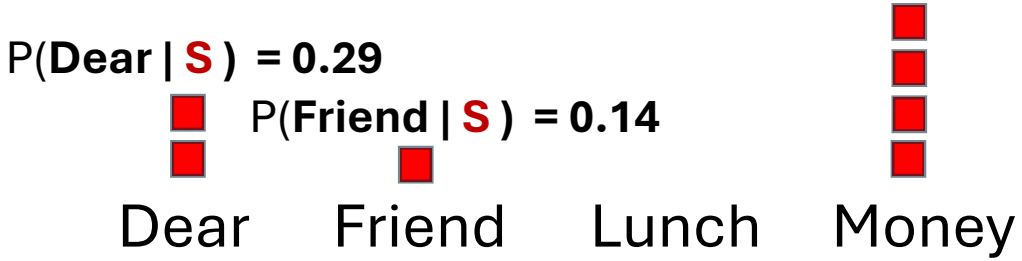


Classification: Naïve Bayes

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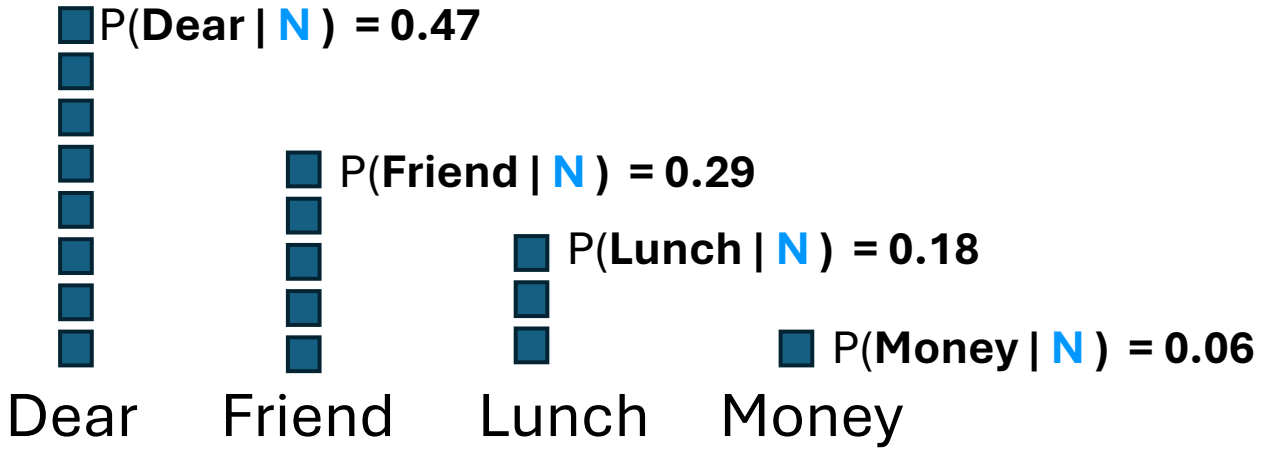


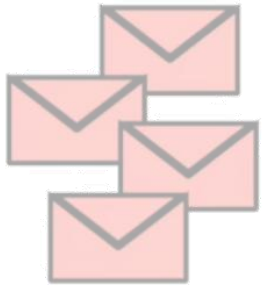


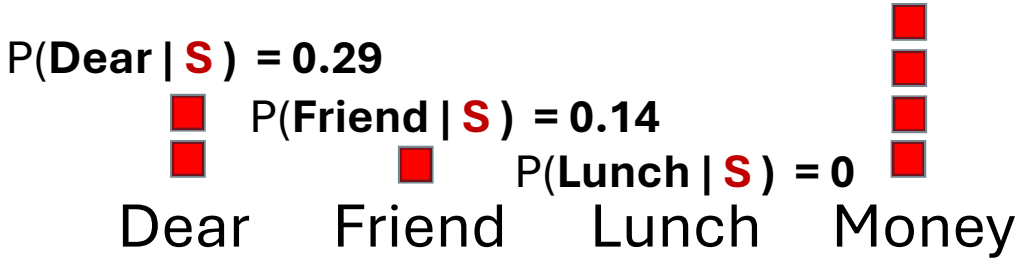


Classification: Naïve Bayes

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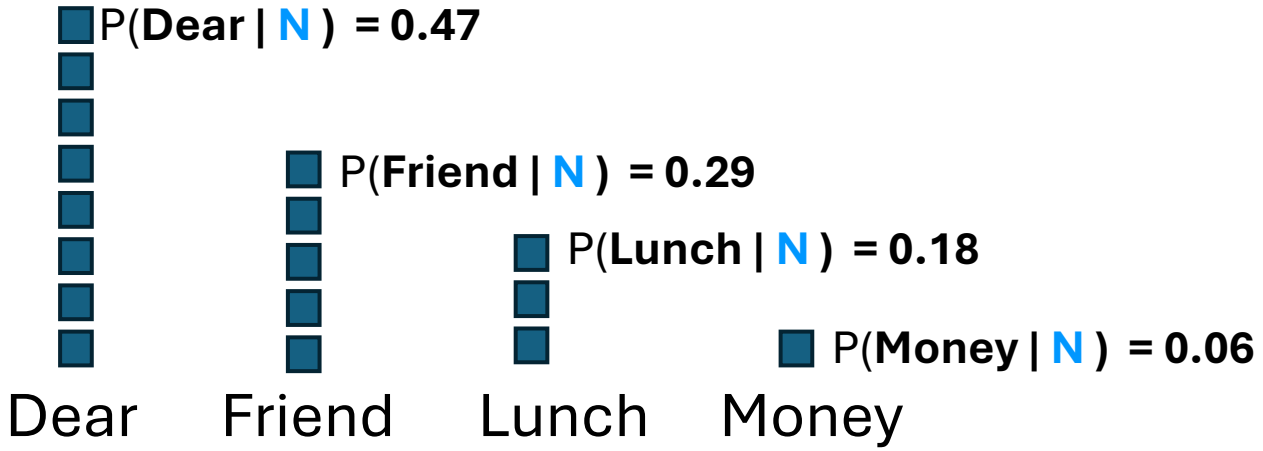


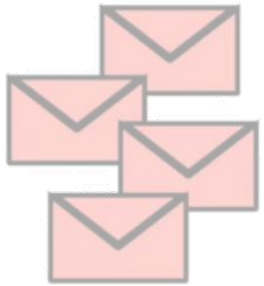


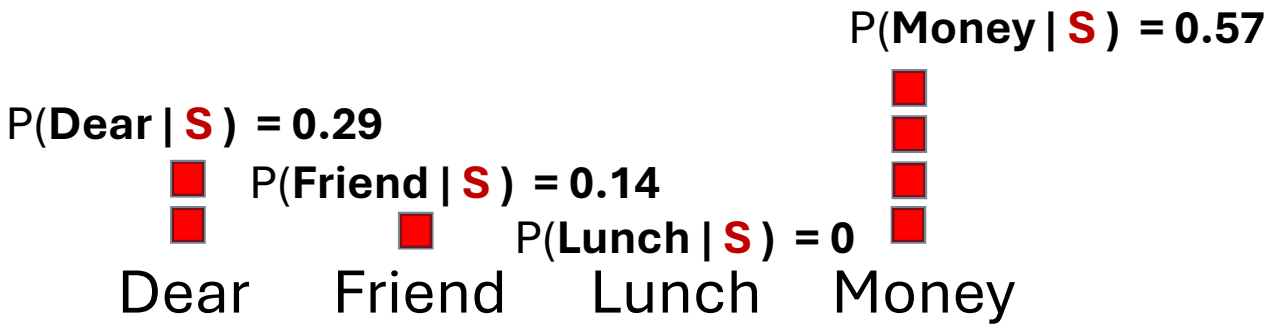


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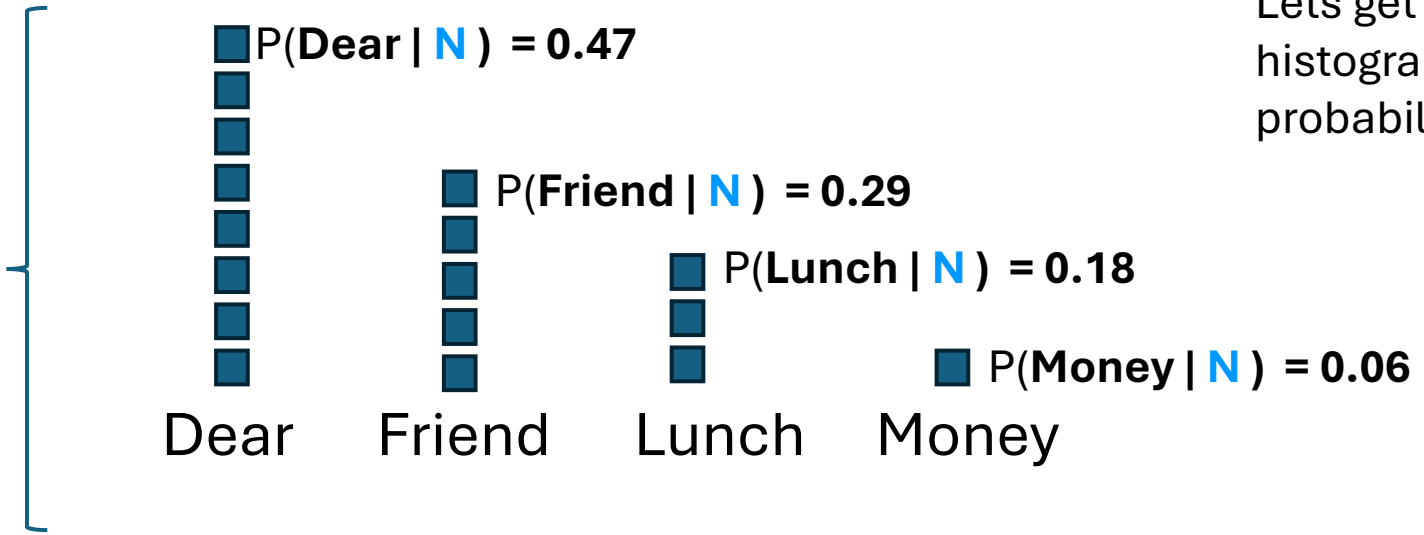


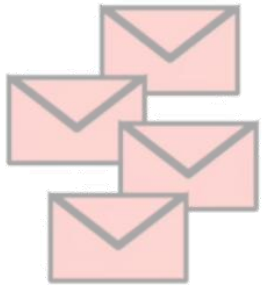


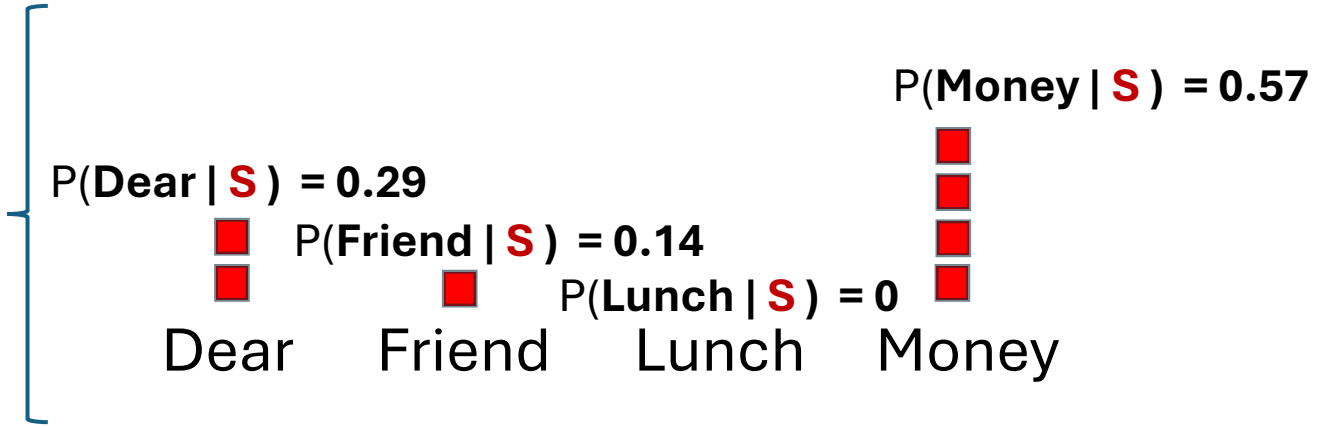


Classification: Naïve Bayes

The Working of Naïve Bayes







Lets get rid of these histograms but keep the probabilities.

Classification: Naïve Bayes

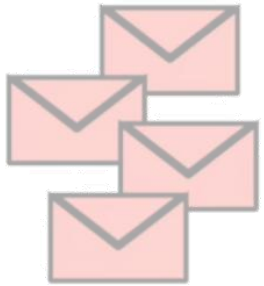
The Working of Naïve Bayes

P(**Dear** | **N**) = **0.47**

P(**Friend** | **N**) = **0.29**

P(**Lunch** | **N**) = **0.18**

P(**Money** | **N**) = **0.06**



P(**Dear** | **S**) = **0.29**

P(**Friend** | **S**) = **0.14**

P(**Lunch** | **S**) = **0**

P(**Money** | **S**) = **0.57**

These probabilities also called ***likelihoods*** because, as we calculated these from discrete individual words

Classification: Naïve Bayes

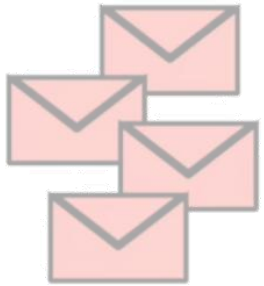
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P(**Dear | S) = 0.29**

P(**Friend | S) = 0.14**

P(**Lunch | S) = 0**

P(**Money | S) = 0.57**

Lets suppose we got a new email saying

“ Dear Friend ”

We start with an initial guess about the probability that any email, regardless of what it says, is a Normal email. This guess will come from training data.

$$p(\mathbf{N}) = \frac{8}{8+4} = 0.67$$

Classification: Naïve Bayes

Lets suppose we got a new email saying
“ Dear Friend ”

The Working of Naïve Bayes

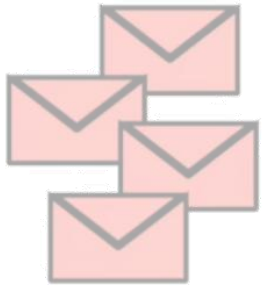
P(**Dear | N**) **= 0.47**

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P(**Lunch | N**) **= 0.18**

P(**Money | N**) **= 0.06**

p(N) = 0.67 This is also called ***Prior Probability***



P(**Dear | S**) **= 0.29**

P(**Friend | S**) **= 0.14**

P(**Lunch | S**) **= 0**

P(**Money | S**) **= 0.57**

Classification: Naïve Bayes

The Working of Naïve Bayes

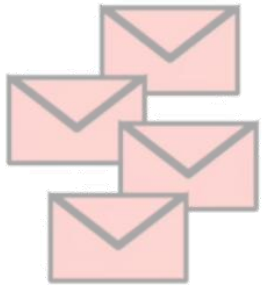
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“ Dear Friend ”

P(**Dear | N) = 0.47**
P(**Friend | N) = 0.29**
P(**Lunch | N) = 0.18**
P(**Money | N) = 0.06**

p(**N**) = 0.67

= p(**N**)x P(**Dear | N)**

Now we multiply the initial guess by the probability that the word ***Dear*** occurs in a normal message



P(**Dear | S) = 0.29**
P(**Friend | S) = 0.14**
P(**Lunch | S) = 0**
P(**Money | S) = 0.57**

Classification: Naïve Bayes

The Working of Naïve Bayes

Lets suppose we got a new email saying
“ Dear Friend ”

P(**Dear | N) = 0.47**

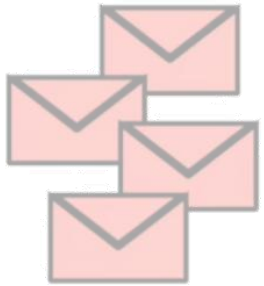
P(**Friend | N) = 0.29**

P(**Lunch | N) = 0.18**

P(**Money | N) = 0.06**

p(**N**) = 0.67

= p(**N**)x P(**Dear | N)** x P(**Friend | N)**



P(**Dear | S) = 0.29**

P(**Friend | S) = 0.14**

P(**Lunch | S) = 0**

P(**Money | S) = 0.57**

Classification: Naïve Bayes

The Working of Naïve Bayes

Lets suppose we got a new email saying
“ Dear Friend ”

P(**Dear | N**) = **0.47**

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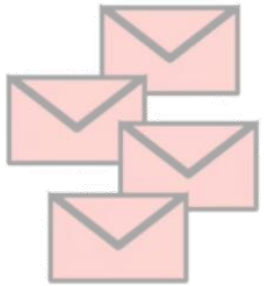
P(**Lunch | N**) = **0.18**

P(**Money | N**) = **0.06**

p(**N**) = 0.67

= p(**N**)x P(**Dear | N**) x P(**Friend | N**)

= 0.67 x 0.47 x 0.29



P(**Dear | S**) = **0.29**

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P(**Lunch | S**) = **0**

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Classification: Naïve Bayes

The Working of Naïve Bayes

Lets suppose we got a new email saying
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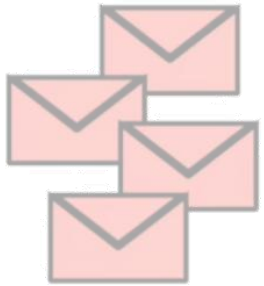
P(**Money | N**) **= 0.06**

p(**N**) = 0.67

= p(**N**)x P(**Dear | N**) x P(**Friend | N**)

= 0.67 x 0.47 x 0.29

=0.09



P(**Dear | S**) **= 0.29**

P(**Friend | S**) **= 0.14**

P(**Lunch | S**) **= 0**

P(**Money | S**) **= 0.57**

We can think of 0.09 as the score that ***Dear Friend*** gets if it is a normal email.

However, technically, it is proportional to the probability that the message is normal given its says ***Dear Friend.***

Classification: Naïve Bayes

The Working of Naïve Bayes

p(**N | Dear Friend)** α 0.09

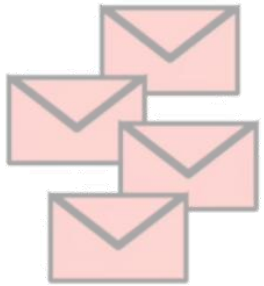
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p(**N**) = 0.67



P(**Dear | S) = 0.29**

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P(**Money | S) = 0.57**

Lets suppose we got a new email saying
“ Dear Friend ”

Now, just like we did before, We start with an initial guess about the probability that any email, regardless of what it says, is a spam.

$$p(\mathbf{S}) = \frac{4}{8 + 4} = 0.33$$

Classification: Naïve Bayes

The Working of Naïve Bayes

p(**N** | **Dear Friend**) α 0.09

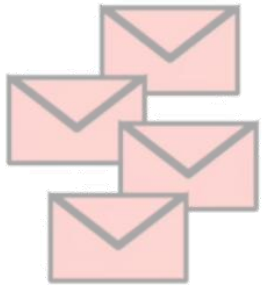
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P(**Dear** | **S**) = **0.29**

P(**Friend** | **S**) = **0.14**

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P(**Money** | **S**) = **0.57**

p(**S**) = 0.33

Lets suppose we got a new email saying
“ Dear Friend ”

Now, just like we did before, We start with an initial guess about the probability that any email, regardless of what it says, is a spam.

= p(**S**)x P(**Dear** | **S**) x P(**Friend** | **S**)

Classification: Naïve Bayes

The Working of Naïve Bayes

p(**N | Dear Friend)** α 0.09

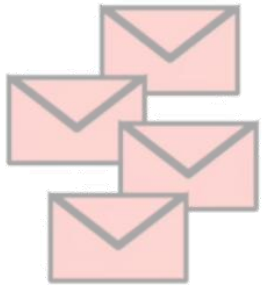
P(**Dear | N) = 0.47**

P(**Friend | N) = 0.29**

P(**Lunch | N) = 0.18**

P(**Money | N) = 0.06**

p(**N**) = 0.67



P(**Dear | S) = 0.29**

P(**Friend | S) = 0.14**

P(**Lunch | S) = 0**

P(**Money | S) = 0.57**

p(**S**) = 0.33

Lets suppose we got a new email saying
“ Dear Friend ”

Now, just like we did before, We start with an initial guess about the probability that any email, regardless of what it says, is a spam.

= p(**S**)x P(**Dear | S) x P(Friend | S)**
= 0.33 x 0.29 x 0.14

Classification: Naïve Bayes

The Working of Naïve Bayes

p(**N | Dear Friend)** α 0.09

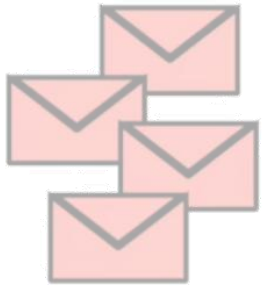
P(**Dear | N) = 0.47**

P(**Friend | N) = 0.29**

P(**Lunch | N) = 0.18**

P(**Money | N) = 0.06**

p(**N**) = 0.67



P(**Dear | S) = 0.29**

P(**Friend | S) = 0.14**

P(**Lunch | S) = 0**

P(**Money | S) = 0.57**

p(**S**) = 0.33

Lets suppose we got a new email saying
“ Dear Friend ”

However, technically, it is proportional to the probability that the message is spam given its says ***Dear Friend***

= p(**S**)x P(**Dear | S) x P(Friend | S)**

= 0.33 x 0.29 x 0.14

= 0.01

= p(**S | Dear Friend)** α 0.01

Classification: Naïve Bayes

The Working of Naïve Bayes

p(**N** | **Dear Friend**) α 0.09

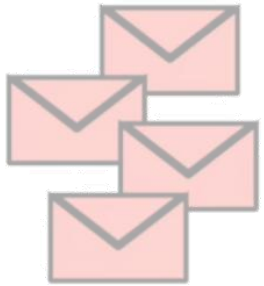
P(**Dear** | **N**) = **0.47**

P(**Friend** | **N**) = **0.29**

P(**Lunch** | **N**) = **0.18**

P(**Money** | **N**) = **0.06**

p(**N**) = 0.67



p(**S** | **Dear Friend**) α 0.01

P(**Dear** | **S**) = **0.29**

P(**Friend** | **S**) = **0.14**

P(**Lunch** | **S**) = **0**

P(**Money** | **S**) = **0.57**

p(**S**) = 0.33

Classification: Naïve Bayes

The Working of Naïve Bayes

p(**N | Dear Friend)** α 0.09

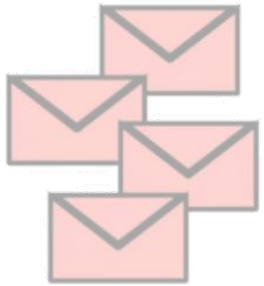
P(**Dear | N) = 0.47**

P(**Friend | N) = 0.29**

P(**Lunch | N) = 0.18**

P(**Money | N) = 0.06**

p(**N**) = 0.67



p(**S | Dear Friend)** α 0.01

P(**Dear | S) = 0.29**

P(**Friend | S) = 0.14**

P(**Lunch | S) = 0**

P(**Money | S) = 0.57**

p(**S**) = 0.33

Hence, the normal score we got for Normal email is 0.09, while for the spam email it is 0.01, means 0.09 is bigger than 0.01…

we will decide that “Dear Friend” is a normal email.

Classification: Naïve Bayes

Lunch Money Money Money Money

The Working of Naïve Bayes

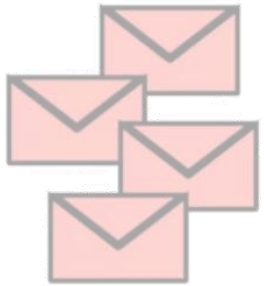
P(**Dear** | **N**) = **0.47**

P(**Friend** | **N**) = **0.29**

P(**Lunch** | **N**) = **0.18**

P(**Money** | **N**) = **0.06** ←

p(**N**) = 0.67



P(**Dear** | **S**) = **0.29**

P(**Friend** | **S**) = **0.14**

P(**Lunch** | **S**) = **0**

P(**Money** | **S**) = **0.57** ←

p(**S**) = 0.33

The probability of seeing the word Money is much higher in spam (0.56) than in normal emails (0.06)

Classification: Naïve Bayes

The Working of Naïve Bayes

Lunch Money Money Money Money

P(**Dear | N**) = **0.47**

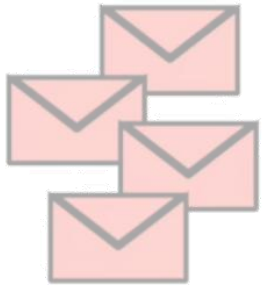
P(**Friend | N**) = **0.29**

P(**Lunch | N**) = **0.18**

P(**Money | N**) = **0.06**

p(**N**) = 0.67

= p(**N**) x P(**Lunch | N**)



P(**Dear | S**) = **0.29**

P(**Friend | S**) = **0.14**

P(**Lunch | S**) = **0**

P(**Money | S**) = **0.57**

p(**S**) = 0.33

Classification: Naïve Bayes

The Working of Naïve Bayes

Lunch Money Money Money Money

P(**Dear | N) = 0.47**

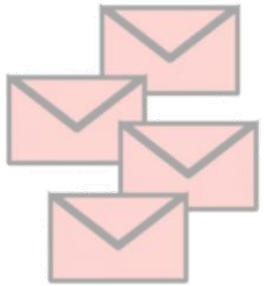
P(**Friend | N) = 0.29**

P(**Lunch | N) = 0.18**

P(**Money | N) = 0.06**

p(**N**) = 0.67

= p(**N**) x P(**Lunch | N) x** p (**Money | N**)^4



P(**Dear | S) = 0.29**

P(**Friend | S) = 0.14**

P(**Lunch | S) = 0**

P(**Money | S) = 0.57**

p(**S**) = 0.33

Classification: Naïve Bayes

The Working of Naïve Bayes

Lunch Money Money Money Money

P(**Dear | N) = 0.47**

P(**Friend | N) = 0.29**

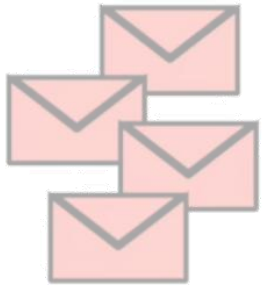
P(**Lunch | N) = 0.18**

P(**Money | N) = 0.06**

p(**N**) = 0.67

= p(**N**) x P(**Lunch | N) x** p (**Money | N**)^4

= 0.000002



P(**Dear | S) = 0.29**

P(**Friend | S) = 0.14**

P(**Lunch | S) = 0**

P(**Money | S) = 0.57**

p(**S**) = 0.33

Classification: Naïve Bayes

Lunch Money Money Money Money

The Working of Naïve Bayes

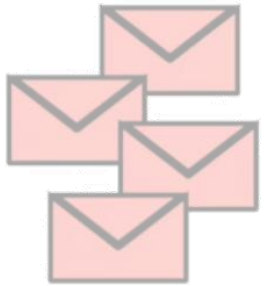
P(**Dear | N**) = **0.47**

P(**Friend | N**) = **0.29**

P(**Lunch | N**) = **0.18**

P(**Money | N**) = **0.06**

p(**N**) = 0.67



P(**Dear | S**) = **0.29**

P(**Friend | S**) = **0.14**

P(**Lunch | S**) = **0**

P(**Money | S**) = **0.57**

p(**S**) = 0.33

= p(**S**) x P(**Lunch | S**) **x** p (**Money | S**)^4

= 0

Classification: Naïve Bayes

Lunch Money Money Money Money

The Working of Naïve Bayes

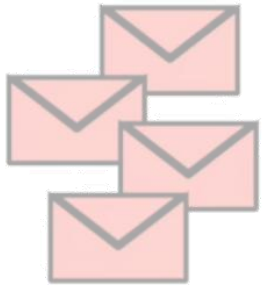
P(**Dear | N**) = **0.47**

P(**Friend | N**) = **0.29**

P(**Lunch | N**) = **0.18**

P(**Money | N**) = **0.06**

p(**N**) = 0.67



P(**Dear | S**) = **0.29**

P(**Friend | S**) = **0.14**

P(**Lunch | S**) = **0**

P(**Money | S**) = **0.57**

p(**S**) = 0.33

= p(**S**) x P(**Lunch | S**) **x** p (**Money | S**)^4

= 0

Classification: Naïve Bayes

The Working of Naïve Bayes

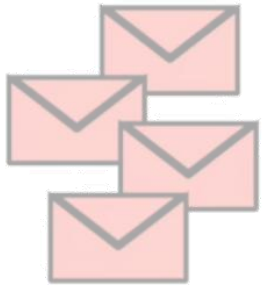
P(**Dear | N**) **= 0.47**

P(**Friend | N**) **= 0.29**

P(**Lunch | N**) **= 0.18**

P(**Money | N**) **= 0.06**

p(**N**) = 0.67



P(**Dear | S**) **= 0.29**

P(**Friend | S**) **= 0.14**

P(**Lunch | S**) **= 0**

P(**Money | S**) **= 0.57**

p(**S**) = 0.33

Lunch Money Money Money Money

Now we have a problem: an email having world Lunch will always be considered Normal email, regardless of the other part of the email.

= p(**N**) x P(**Lunch | N**) **x** p (**Money | N**)^4 = 0.000002

= p(**S**) x P(**Lunch | S**) **x** p (**Money | S**)^4 = 0

Classification: Naïve Bayes

The Working of Naïve Bayes

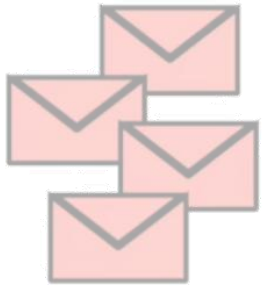
P(**Dear** | **N**) = **0.47**

P(**Friend** | **N**) = **0.29**

P(**Lunch** | **N**) = **0.18**

P(**Money** | **N**) = **0.06**

p(**N**) = 0.67



P(**Dear** | **S**) = **0.29**

P(**Friend** | **S**) = **0.14**

P(**Lunch** | **S**) = **0**

P(**Money** | **S**) = **0.57**

p(**S**) = 0.33

Lunch Money Money Money Money

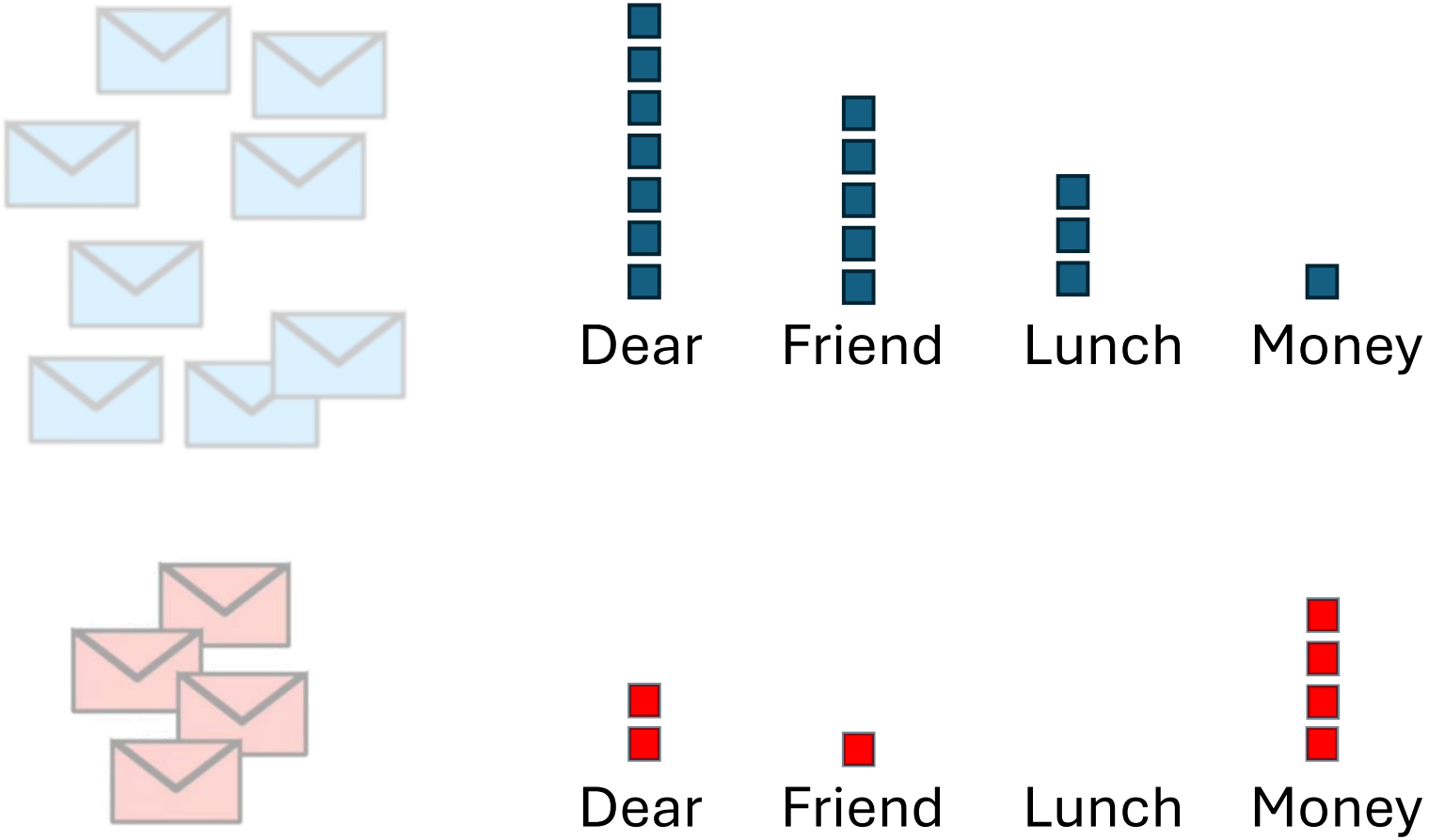
To work around this problem, we will add 1 count in the histogram of each word.

Classification: Naïve Bayes

The Working of Naïve Bayes

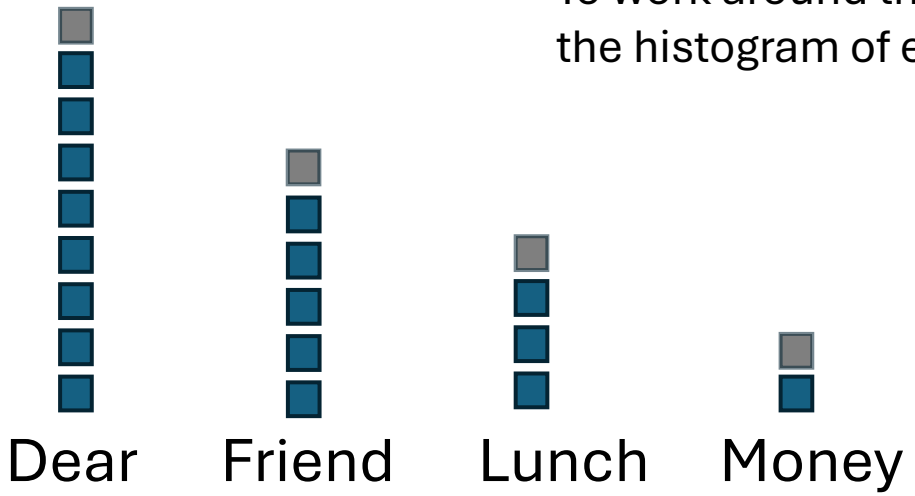
Lunch Money Money Money Money

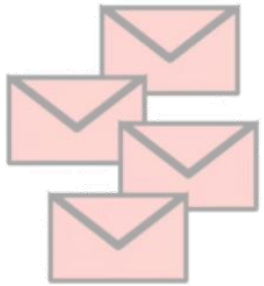
To work around this problem, we will add 1 count in the histogram of each word.

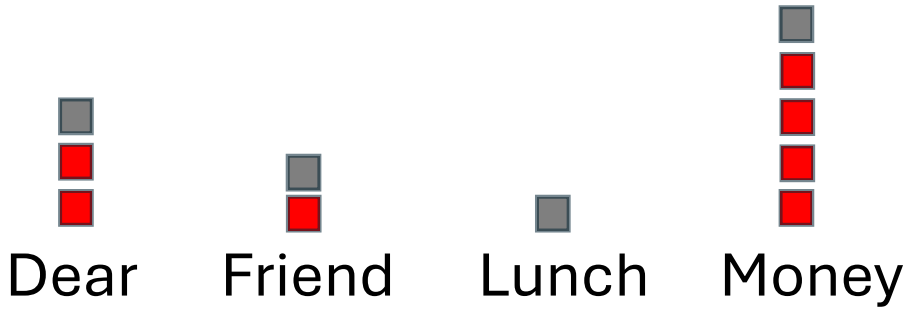


Classification: Naïve Bayes

The Working of Naïve Bayes







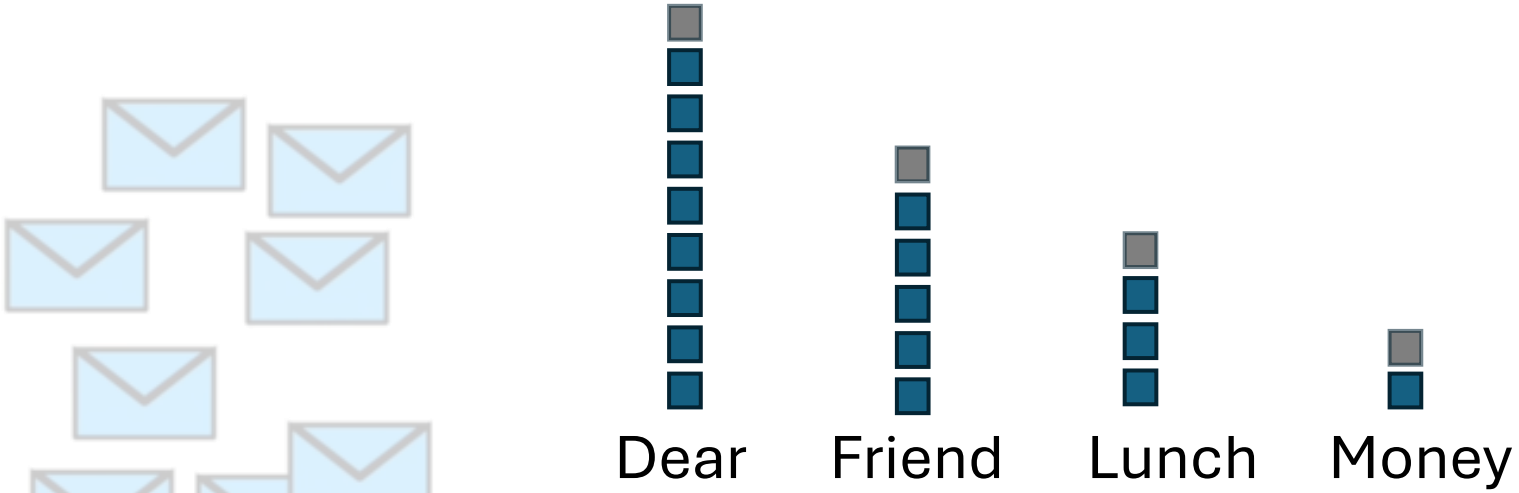
Lunch Money Money Money Money

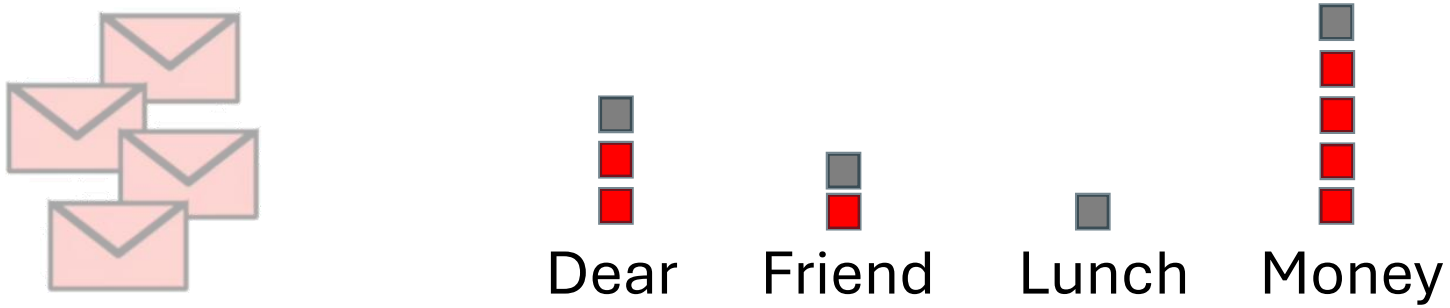
To work around this problem, we will add 1 count in the histogram of each word.

Classification: Naïve Bayes

Lunch Money Money Money Money

<u>The Working of Naïve Bayes</u>





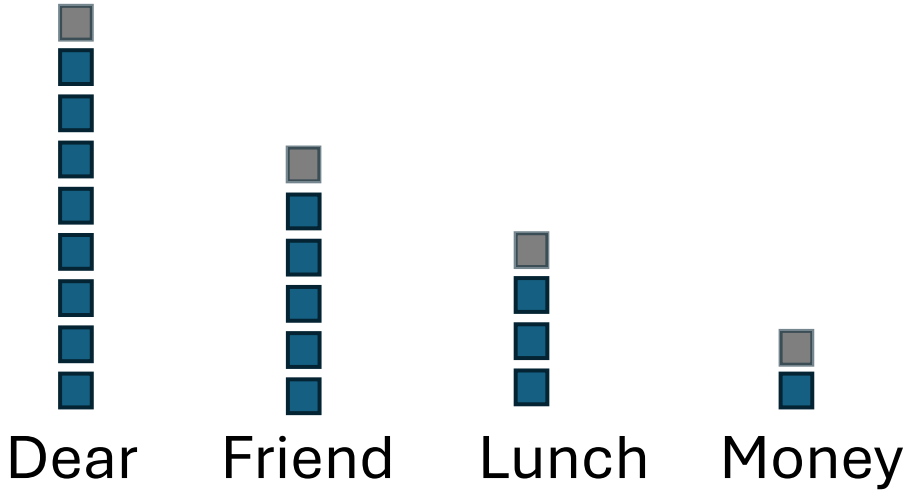
Now the probability of seeing Lunch, given that it is in spam…

= p(**Lunch | S) = 1/7+4**
= 0.09

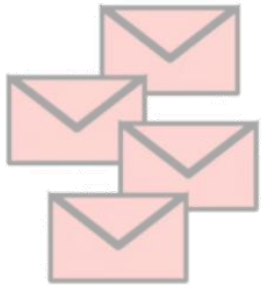
Classification: Naïve Bayes

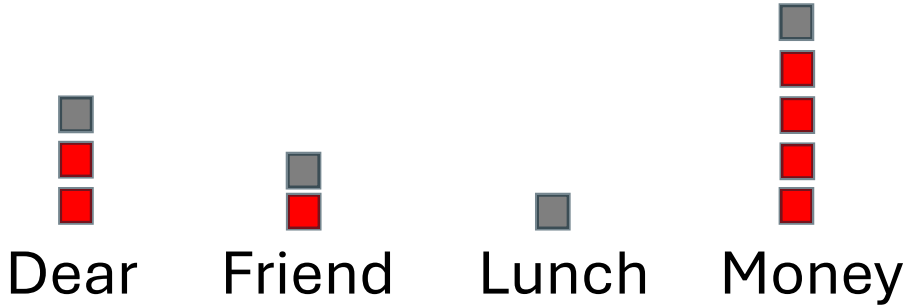
The Working of Naïve Bayes

Lunch Money Money Money Money



Note: adding counts to each word does not change our initial guess that a message is normal p(N)





Classification: Naïve Bayes

The Working of Naïve Bayes

Lunch Money Money Money Money

P(**Dear** | **N**) = **0.43**

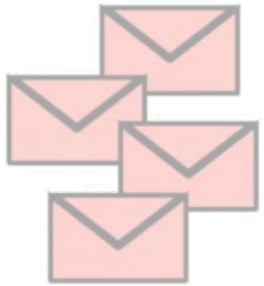
P(**Friend** | **N**) = **0.29**

P(**Lunch** | **N**) = **0.19**

P(**Money** | **N**) = **0.10**

p(**N**) = 0.67

= p(**N**) x P(**Lunch** | **N**) **x** p (**Money** | **N**)^4 = 0.000001



P(**Dear** | **S**) = **0.27**

P(**Friend** | **S**) = **0.18**

P(**Lunch** | **S**) = **0.09**

P(**Money** | **S**) = **0.45**

p(**S**) = 0.33

= p(**S**) x P(**Lunch** | **S**) **x** p (**Money** | **S**)^4 = 0.00122

Classification: Naïve Bayes

Our Question For Today?

The scenario

- A property assessor must place every house in a ***price band*** before the valuation is finalized.
- They have the full record: size, quality, year built, garage, everything.
- They do not yet have the price.

The question?

Given everything else about the house, is it **above** or **below** the market median?

How Naive Bayes asks it differently?

- A tree asked: which yes/no questions separate the two groups?
- Naive Bayes asks: **which price band was more likely to produce a house like this?**

Classification: k-Nearest Neighbors (k-NN)

Definition:

A nearest-neighbor classifier represents each example as a point in space, and classifies a new example by the majority label among its k closest training examples.

Chapter 4
Introduction to Data Mining, 2nd Edition



Bookworms



Soccer kids



Card players

Which group will the new kid join?



New kid

Classification: k-Nearest Neighbors (k-NN)

Where it can be used?

- **Recommendation:** people similar to you also liked this
- **Image and document similarity search:** find the nearest match in a library
- **Credit scoring:** find past applicants most like this one
- **Anomaly detection baseline:** a point with no close neighbors is suspicious

Pros

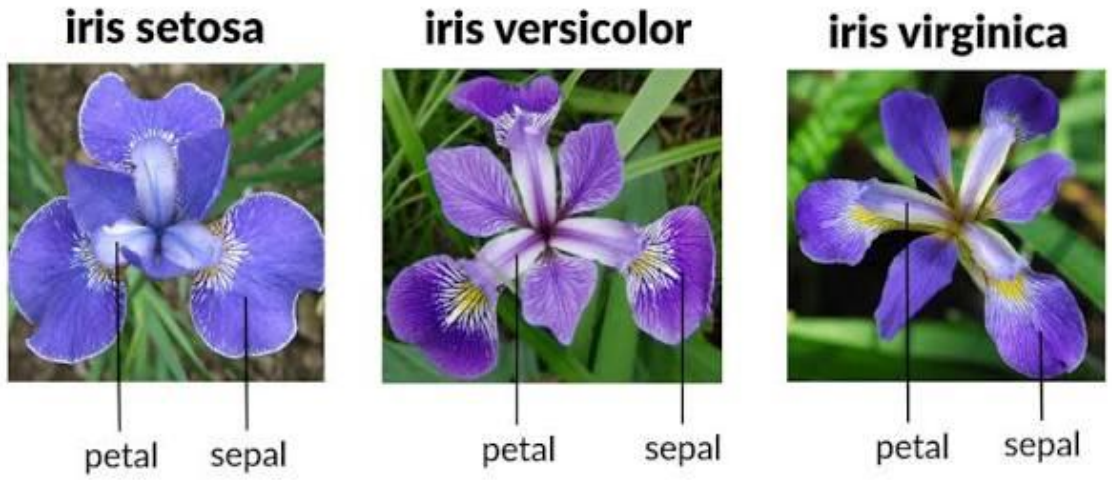
- **Easy to implement:** No training step at all. Store the data and you are done
- **One parameter:** Just k, plus the distance measure
- **Very adaptable:** Works for classification and regression, any number of classes

Cons

- **Scalability:** Every prediction compares against every stored row. Slow on large data
- **Curse of dimensionality:** In high dimensions all distances converge and "nearest" stops meaning anything
- **Needs scaling:** Columns with big numbers dominate the distance

Classification: k-Nearest Neighbors (k-NN)

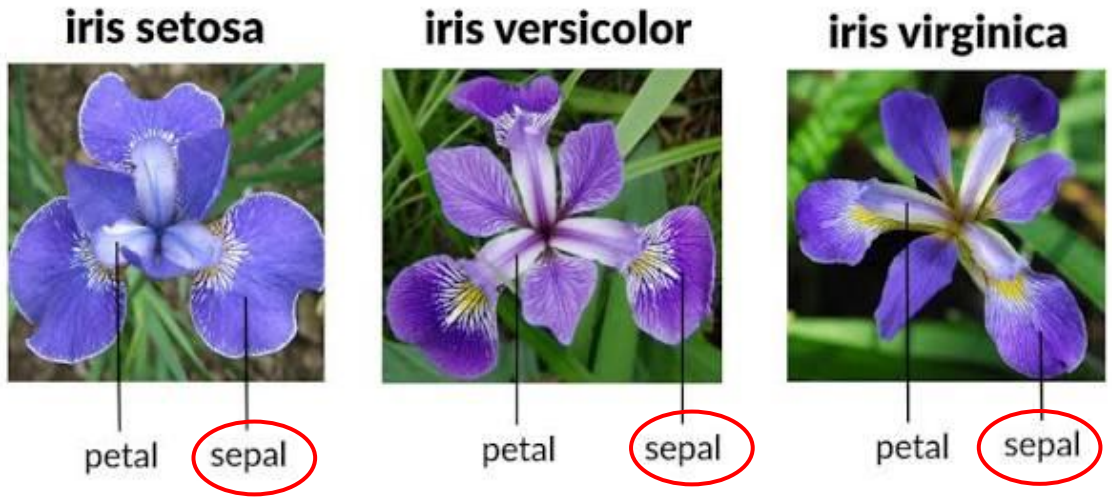
The Working of k-NN



Iris flower data set

Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN



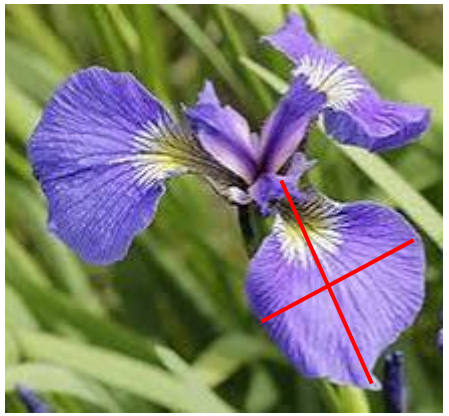
Iris flower data set

Sepal Length	Sepal Width	Species
5.2	3.6	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor

Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species
5.2	3.6	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor



Sepal Length is 5.4
Sepal Width is 3.11

Which Species?

Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species
5.2	3.6	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor
5.2	3.06	**???**

Step 1: Find distance of each sample from the one we need to predict

Euclidean Distance $\sqrt{\sum_{i=1}^{k}(x_i - y_i)^2}$

Manhattan Distance $\sum_{i=1}^{k}|x_i - y_i|$

Minkowski Distance $\left(\sum_{i=1}^{k}(|x_i - y_i|)^q\right)^{1/q}$

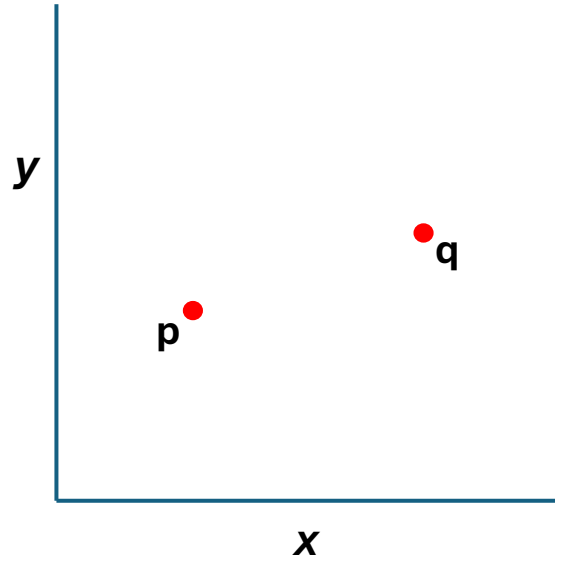
Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species
5.2	3.6	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor
5.2	3.06	**???**

Step 1: Find distance of each sample from the one we need to predict

Euclidean Distance (p,q) $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



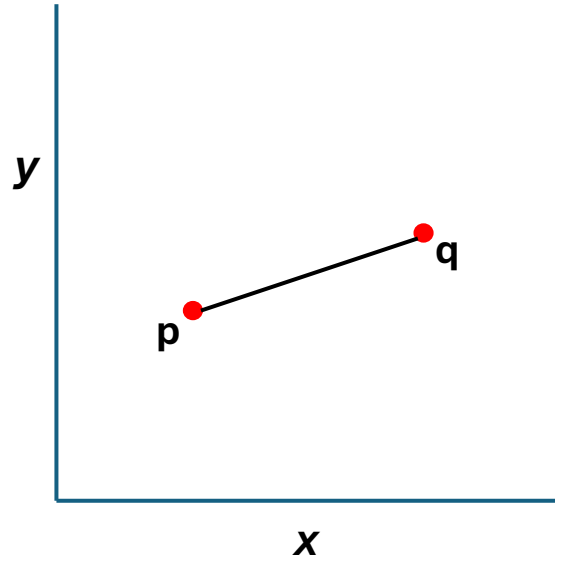
Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species
5.2	3.6	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor
5.2	3.06	**???**

Step 1: Find distance of each sample from the one we need to predict

Euclidean Distance (p,q) $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



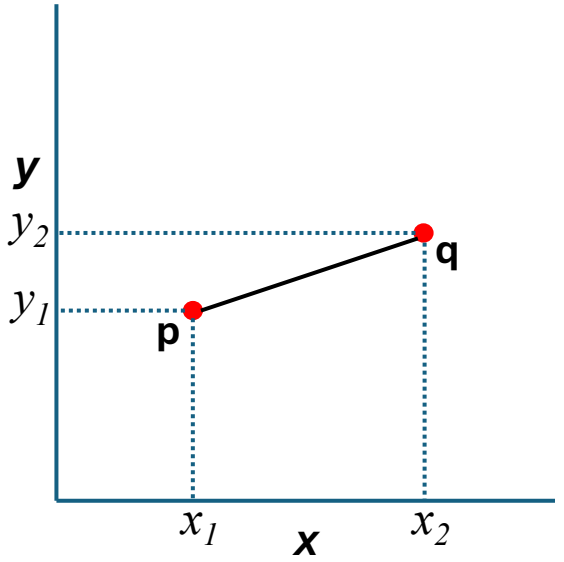
Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species
5.2	3.6	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor
5.2	3.06	**???**

Step 1: Find distance of each sample from the one we need to predict

Euclidean Distance (p,q) $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species
5.2	3.6	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor
5.2	3.06	**???**

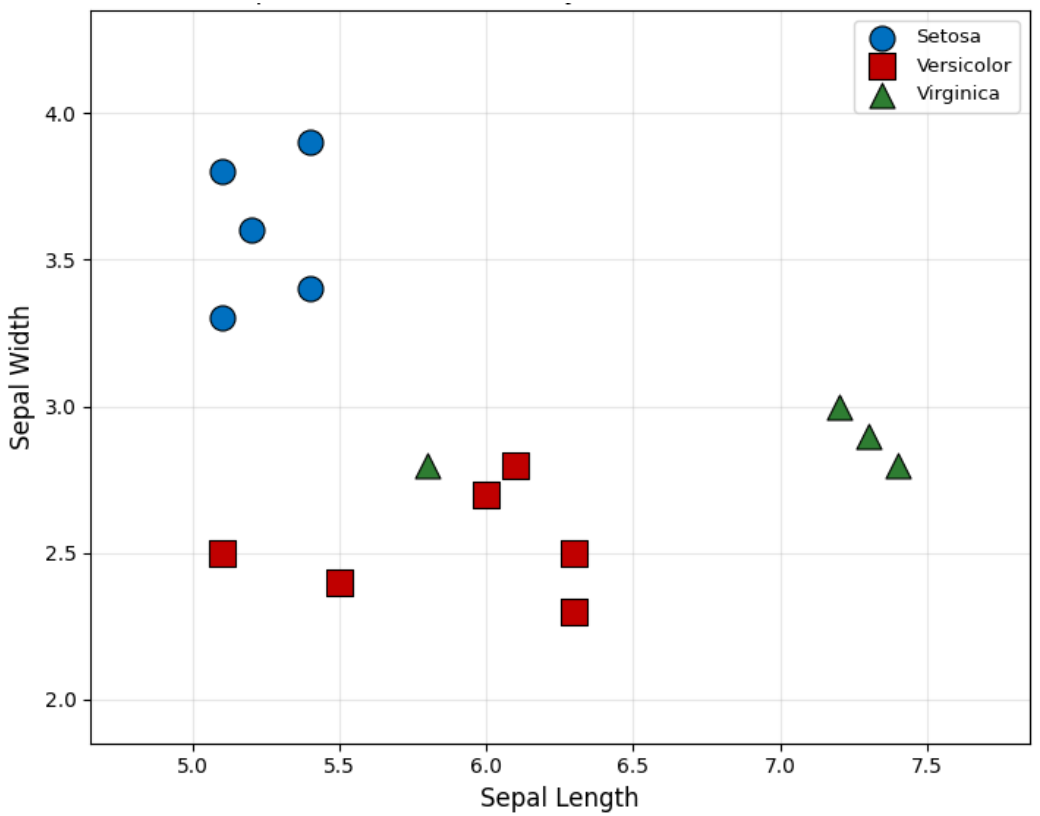
Step 1: Find distance of each sample from the one we need to predict

Step 2: Find the rank of each point from the one we need to predict

Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

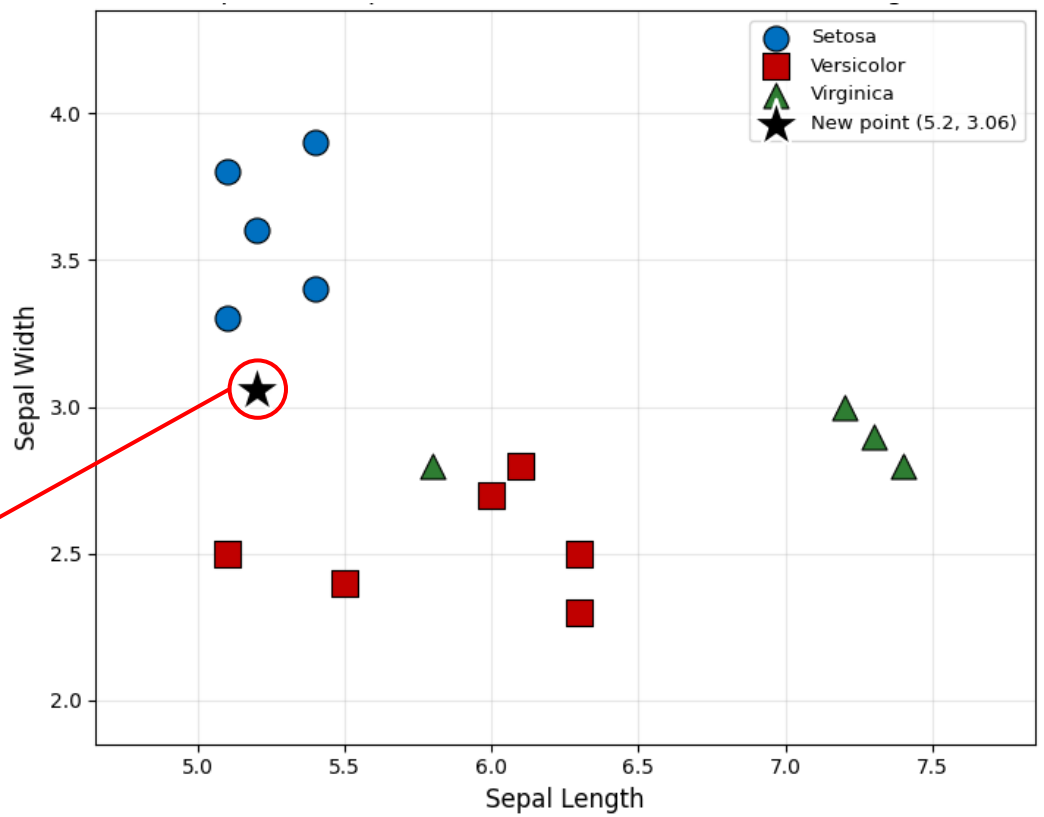
Sepal Length	Sepal Width	Species
5.2	3.6	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor
5.2	3.06	**???**



Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species
5.2	3.6	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor
5.2	3.06	**???**

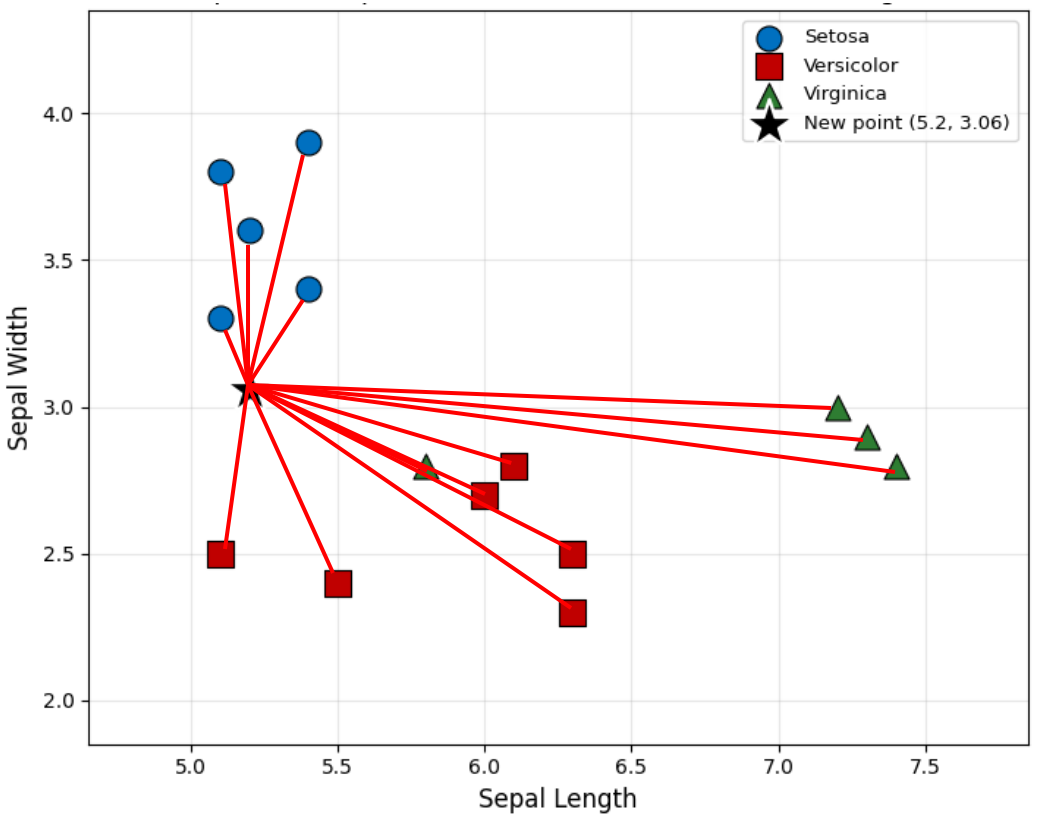


Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species
5.2	3.6	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor
5.2	3.06	**???**

Step 1: Find distance



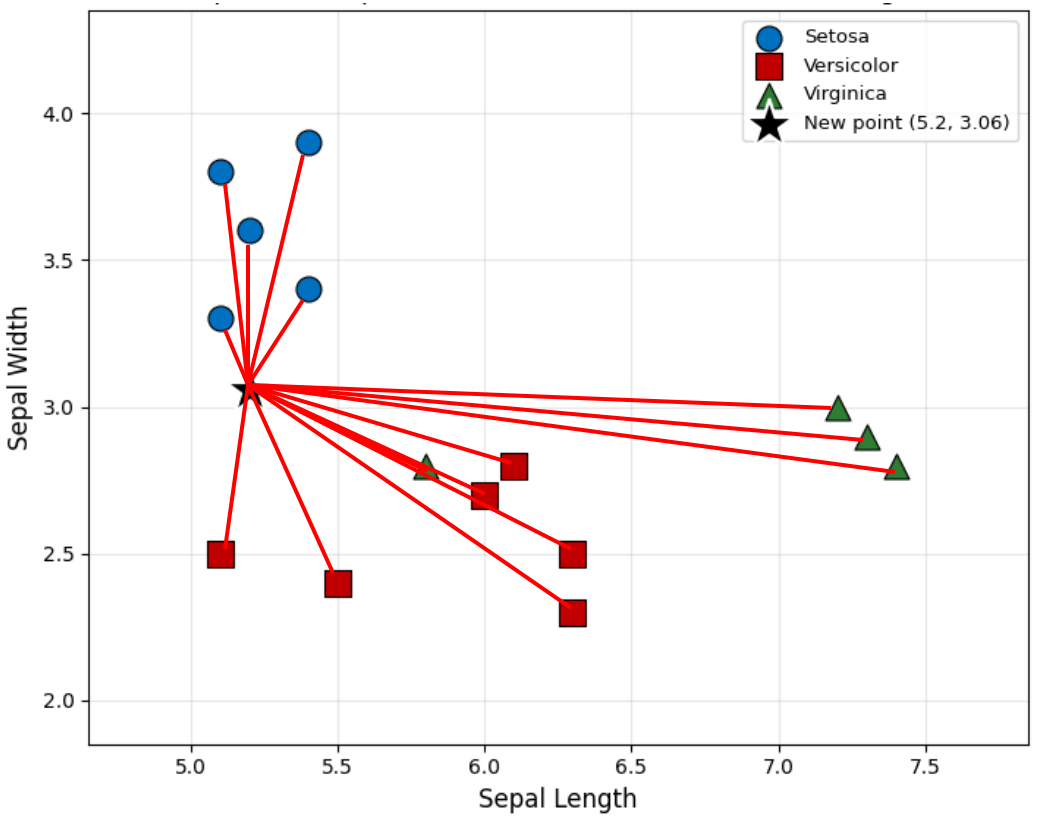
Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species	Distance
5.2	3.6	Setosa	
5.1	3.8	Setosa	
7.2	3.0	Virginica	
5.4	3.4	Setosa	
5.1	3.3	Setosa	
5.4	3.9	Setosa	
7.4	2.8	Virginica	
6.1	2.8	Versicolor	
7.3	2.9	Virginica	
6.0	2.7	Versicolor	
5.8	2.8	Virginica	
6.3	2.3	Versicolor	
5.1	2.5	Versicolor	
6.3	2.5	Versicolor	
5.5	2.4	Versicolor	
5.2	3.06	**???**	

Step 1: Find distance

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)}$$



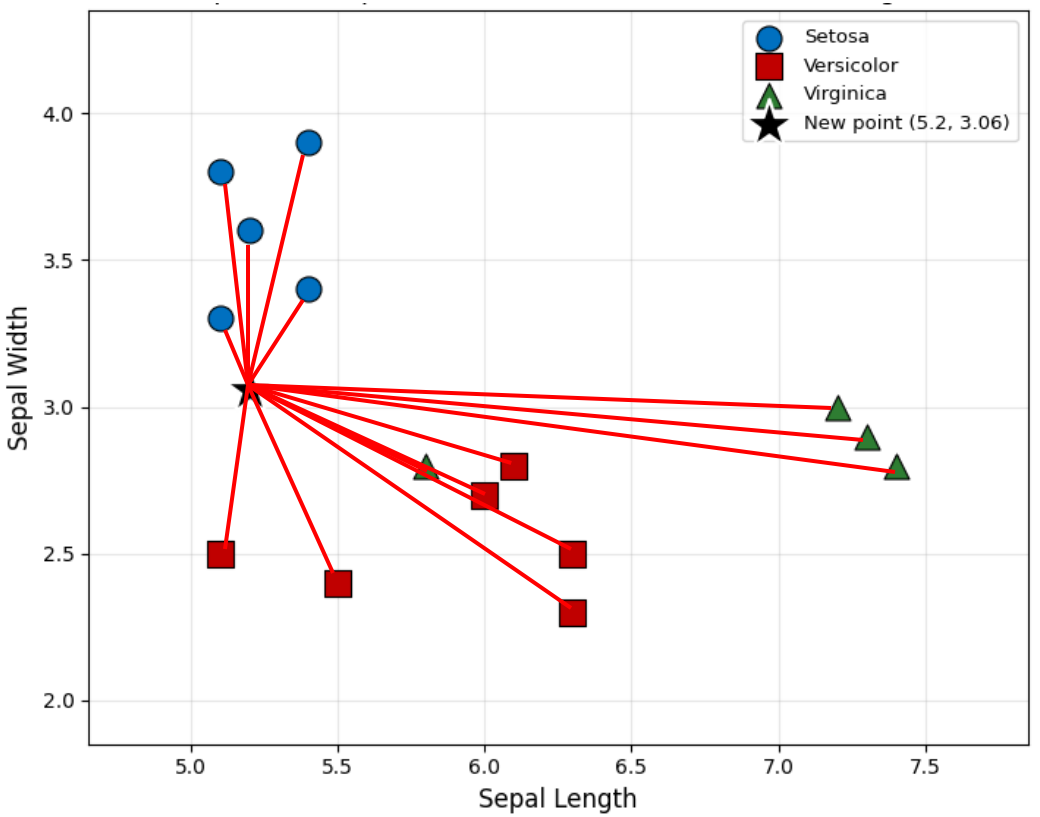
Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species	Distance
5.2	3.6	Setosa	
5.1	3.8	Setosa	
7.2	3.0	Virginica	
5.4	3.4	Setosa	
5.1	3.3	Setosa	
5.4	3.9	Setosa	
7.4	2.8	Virginica	
6.1	2.8	Versicolor	
7.3	2.9	Virginica	
6.0	2.7	Versicolor	
5.8	2.8	Virginica	
6.3	2.3	Versicolor	
5.1	2.5	Versicolor	
6.3	2.5	Versicolor	
5.5	2.4	Versicolor	
5.2	3.06	**???**	

Step 1: Find distance

$$d=\sqrt{(5.2-5.2)^2+(3.6-3.06)}$$



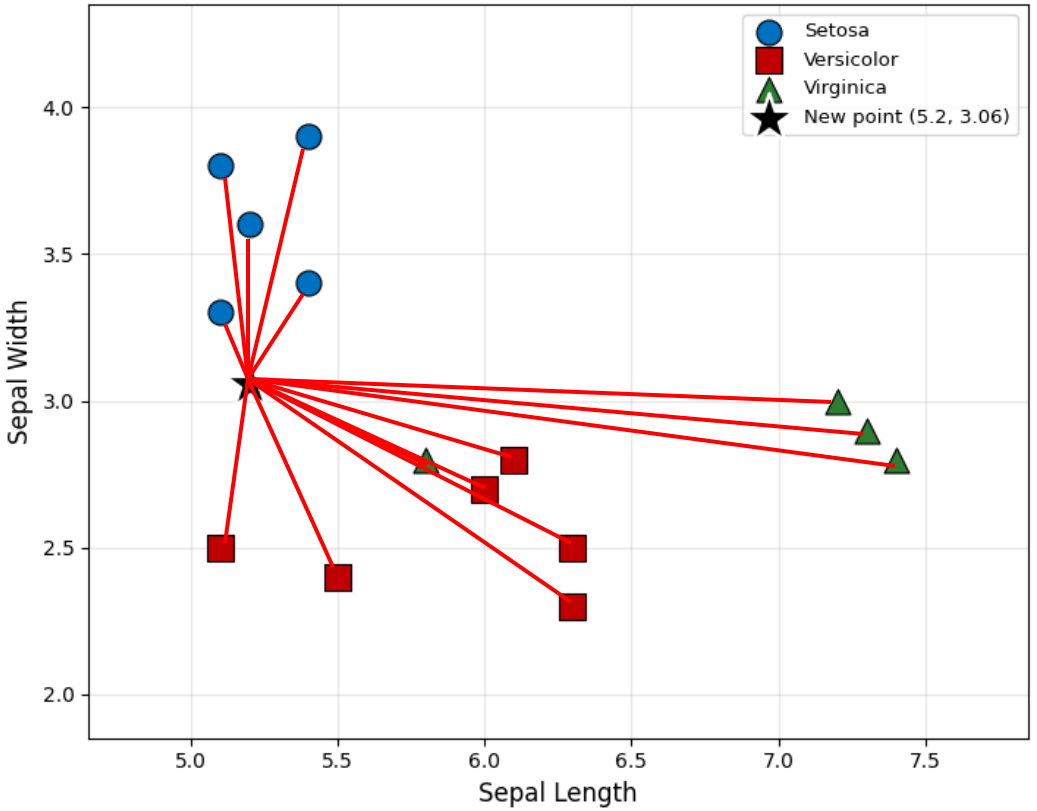
Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species	Distance
5.2	3.6	Setosa	0.540
5.1	3.8	Setosa	
7.2	3.0	Virginica	
5.4	3.4	Setosa	
5.1	3.3	Setosa	
5.4	3.9	Setosa	
7.4	2.8	Virginica	
6.1	2.8	Versicolor	
7.3	2.9	Virginica	
6.0	2.7	Versicolor	
5.8	2.8	Virginica	
6.3	2.3	Versicolor	
5.1	2.5	Versicolor	
6.3	2.5	Versicolor	
5.5	2.4	Versicolor	
5.2	3.06	**???**	

Step 1: Find distance

$$d=\sqrt{(5.2-5.2)^2+(3.6-3.06)} = 0.540$$



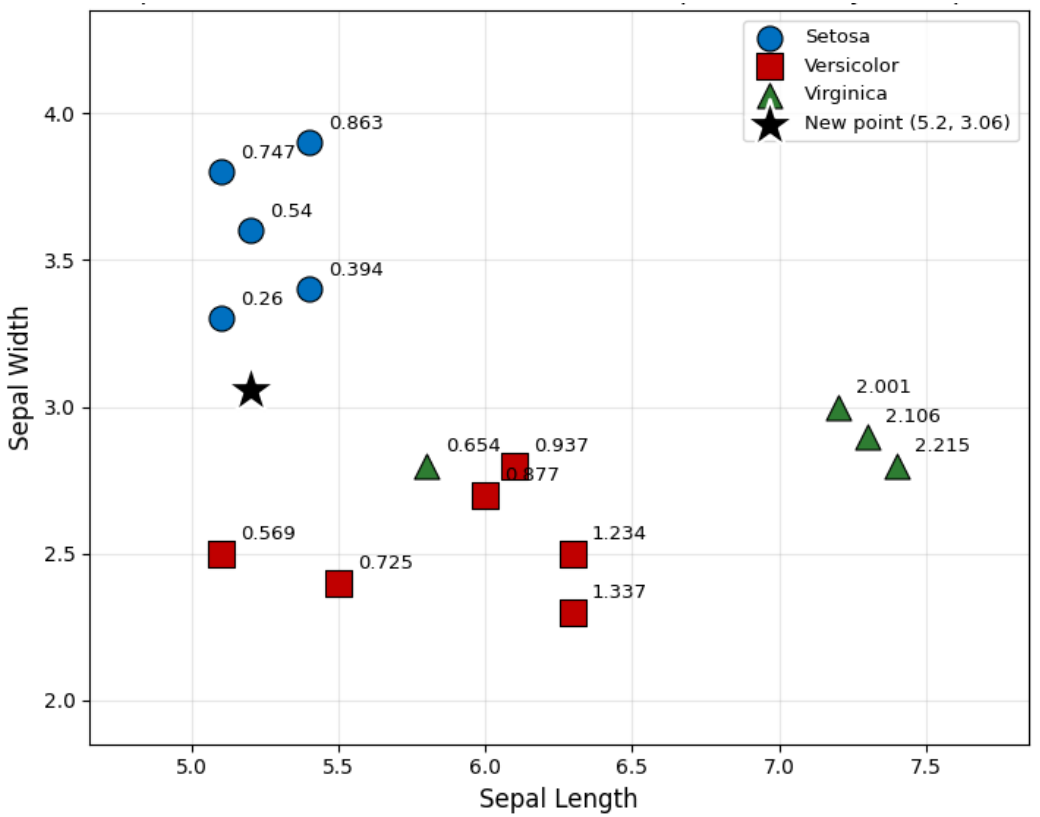
Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species	Distance
5.2	3.6	Setosa	0.540
5.1	3.8	Setosa	0.747
7.2	3.0	Virginica	2.001
5.4	3.4	Setosa	0.394
5.1	3.3	Setosa	0.260
5.4	3.9	Setosa	0.863
7.4	2.8	Virginica	2.215
6.1	2.8	Versicolor	0.937
7.3	2.9	Virginica	2.106
6.0	2.7	Versicolor	0.877
5.8	2.8	Virginica	0.654
6.3	2.3	Versicolor	1.337
5.1	2.5	Versicolor	0.569
6.3	2.5	Versicolor	1.234
5.5	2.4	Versicolor	0.725
5.2	3.06	**???**	

Step 1: Find distance

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)}$$

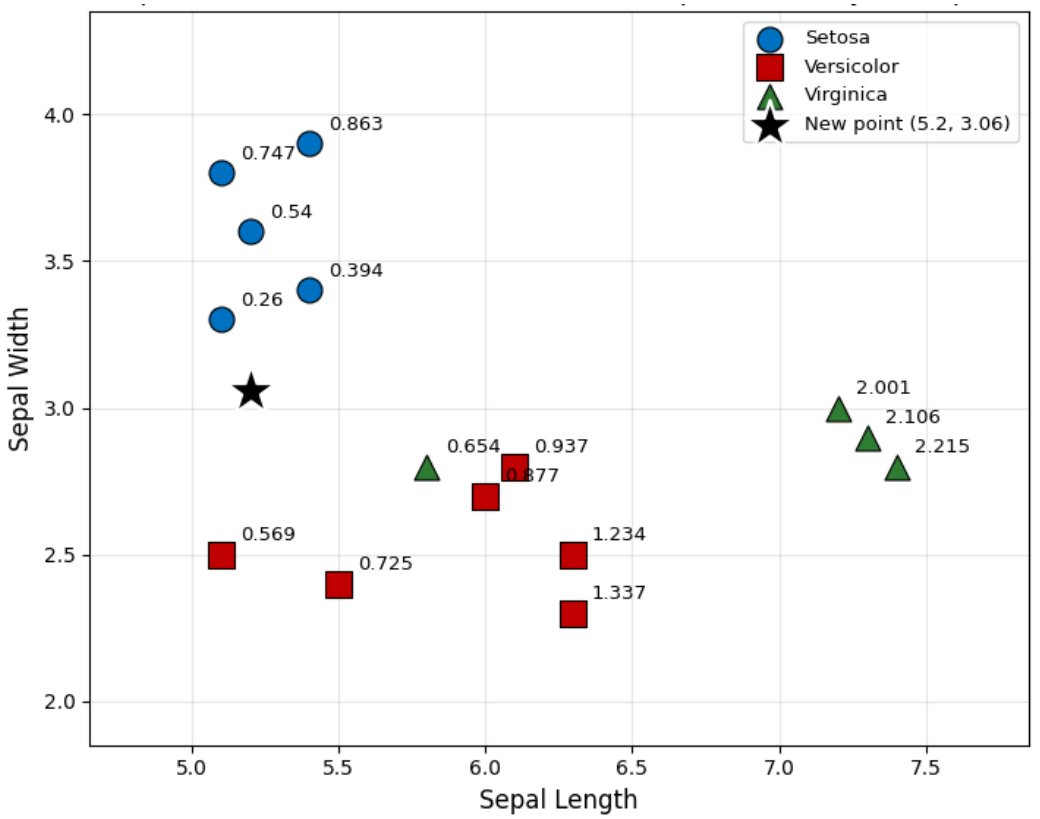


Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species	Distance	Rank
5.2	3.6	Setosa	0.540	3
5.1	3.8	Setosa	0.747	7
7.2	3.0	Virginica	2.001	13
5.4	3.4	Setosa	0.394	2
5.1	3.3	Setosa	0.260	1
5.4	3.9	Setosa	0.863	8
7.4	2.8	Virginica	2.215	15
6.1	2.8	Versicolor	0.937	10
7.3	2.9	Virginica	2.106	14
6.0	2.7	Versicolor	0.877	9
5.8	2.8	Virginica	0.654	5
6.3	2.3	Versicolor	1.337	12
5.1	2.5	Versicolor	0.569	4
6.3	2.5	Versicolor	1.234	11
5.5	2.4	Versicolor	0.725	6
5.2	3.06	**???**		

Step 2: Rank the distances



Classification: k-Nearest Neighbors (k-NN)

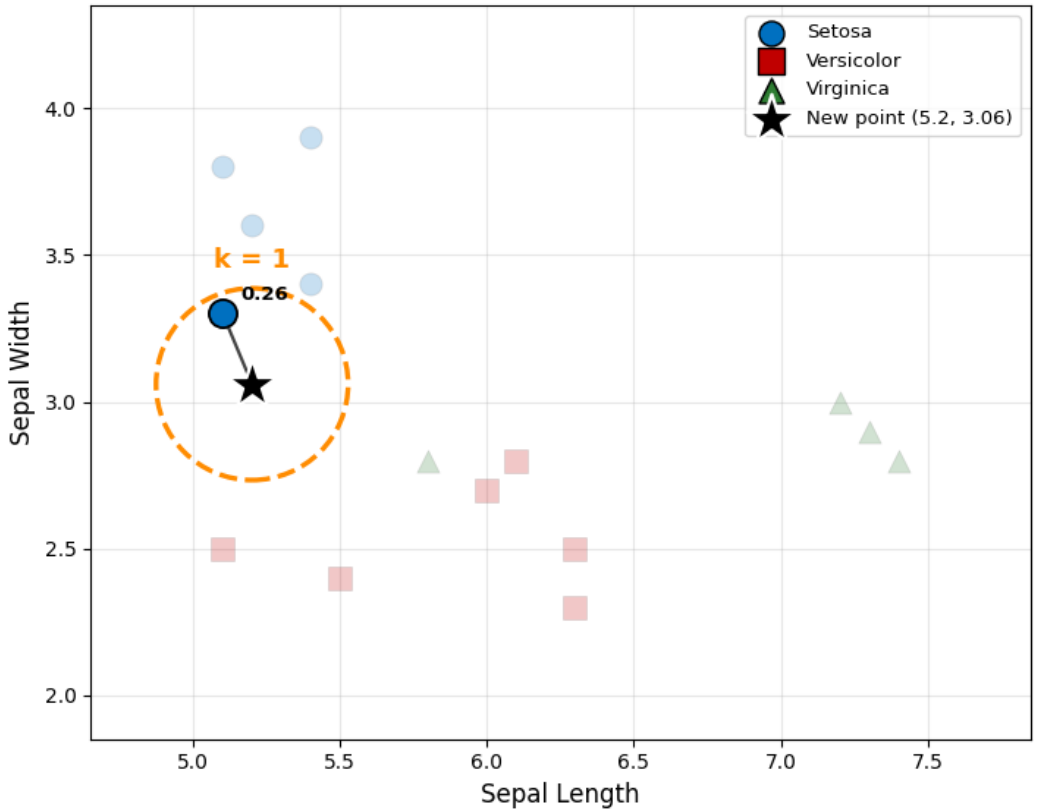
The Working of k-NN

Sepal Length	Sepal Width	Species	Distance	Rank
5.2	3.6	Setosa	0.540	3
5.1	3.8	Setosa	0.747	7
7.2	3.0	Virginica	2.001	13
5.4	3.4	Setosa	0.394	2
5.1	3.3	Setosa	0.260	1
5.4	3.9	Setosa	0.863	8
7.4	2.8	Virginica	2.215	15
6.1	2.8	Versicolor	0.937	10
7.3	2.9	Virginica	2.106	14
6.0	2.7	Versicolor	0.877	9
5.8	2.8	Virginica	0.654	5
6.3	2.3	Versicolor	1.337	12
5.1	2.5	Versicolor	0.569	4
6.3	2.5	Versicolor	1.234	11
5.5	2.4	Versicolor	0.725	6
5.2	3.06	**???**		

Setosa: 01
Virginica: 00
Versicolor: 00

Step 2: Rank the distances

K=1



Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species	Distance	Rank
5.2	3.6	Setosa	0.540	3
5.1	3.8	Setosa	0.747	7
7.2	3.0	Virginica	2.001	13
5.4	3.4	Setosa	0.394	2
5.1	3.3	Setosa	0.260	1
5.4	3.9	Setosa	0.863	8
7.4	2.8	Virginica	2.215	15
6.1	2.8	Versicolor	0.937	10
7.3	2.9	Virginica	2.106	14
6.0	2.7	Versicolor	0.877	9
5.8	2.8	Virginica	0.654	5
6.3	2.3	Versicolor	1.337	12
5.1	2.5	Versicolor	0.569	4
6.3	2.5	Versicolor	1.234	11
5.5	2.4	Versicolor	0.725	6
5.2	3.06	**Setosa**		

Setosa: 01
Virginica: 00
Versicolor: 00

↓

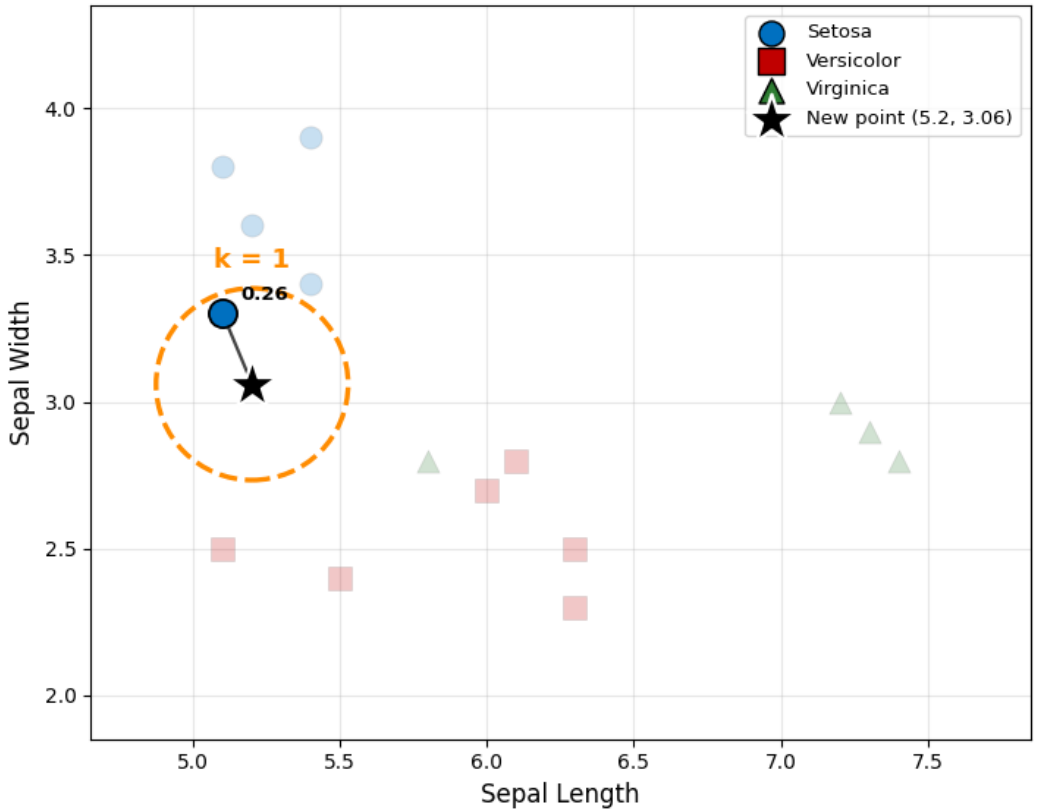
Max

↓

Setosa

Step 2: Rank the distances

K=1



Classification: k-Nearest Neighbors (k-NN)

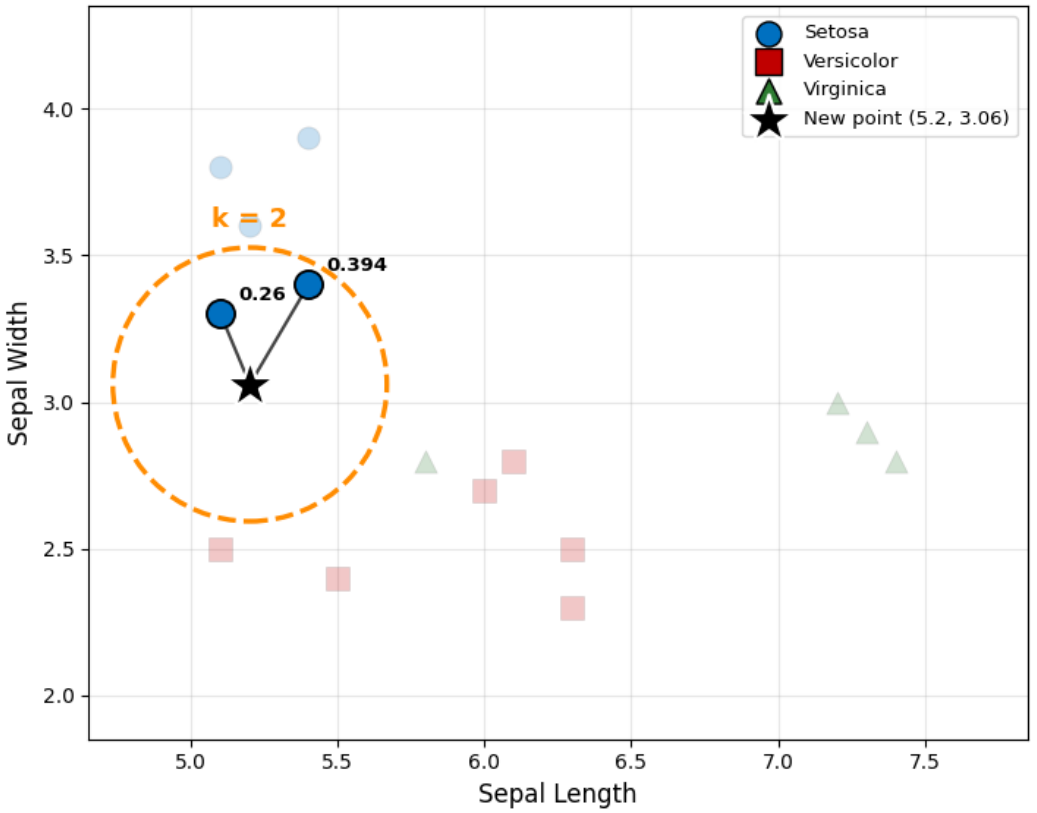
The Working of k-NN

Sepal Length	Sepal Width	Species	Distance	Rank
5.2	3.6	Setosa	0.540	3
5.1	3.8	Setosa	0.747	7
7.2	3.0	Virginica	2.001	13
5.4	3.4	Setosa	0.394	2
5.1	3.3	Setosa	0.260	1
5.4	3.9	Setosa	0.863	8
7.4	2.8	Virginica	2.215	15
6.1	2.8	Versicolor	0.937	10
7.3	2.9	Virginica	2.106	14
6.0	2.7	Versicolor	0.877	9
5.8	2.8	Virginica	0.654	5
6.3	2.3	Versicolor	1.337	12
5.1	2.5	Versicolor	0.569	4
6.3	2.5	Versicolor	1.234	11
5.5	2.4	Versicolor	0.725	6
5.2	3.06	**???**		

Setosa: 02
Virginica: 00
Versicolor: 00

Step 2: Rank the distances

K=2



Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species	Distance	Rank
5.2	3.6	Setosa	0.540	3
5.1	3.8	Setosa	0.747	7
7.2	3.0	Virginica	2.001	13
5.4	3.4	Setosa	0.394	2
5.1	3.3	Setosa	0.260	1
5.4	3.9	Setosa	0.863	8
7.4	2.8	Virginica	2.215	15
6.1	2.8	Versicolor	0.937	10
7.3	2.9	Virginica	2.106	14
6.0	2.7	Versicolor	0.877	9
5.8	2.8	Virginica	0.654	5
6.3	2.3	Versicolor	1.337	12
5.1	2.5	Versicolor	0.569	4
6.3	2.5	Versicolor	1.234	11
5.5	2.4	Versicolor	0.725	6
5.2	3.06	**Setosa**		

Setosa: 02
Virginica: 00
Versicolor: 00

↓

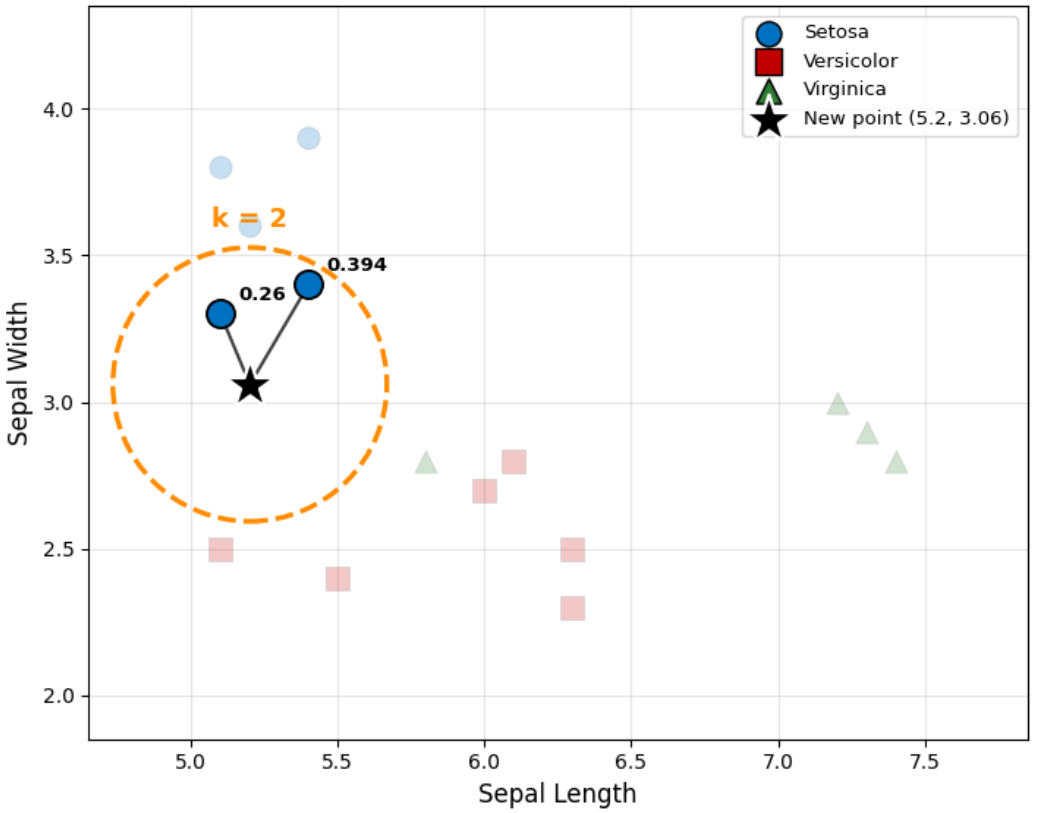
Max

↓

Setosa

Step 2: Rank the distances

K=2



Classification: k-Nearest Neighbors (k-NN)

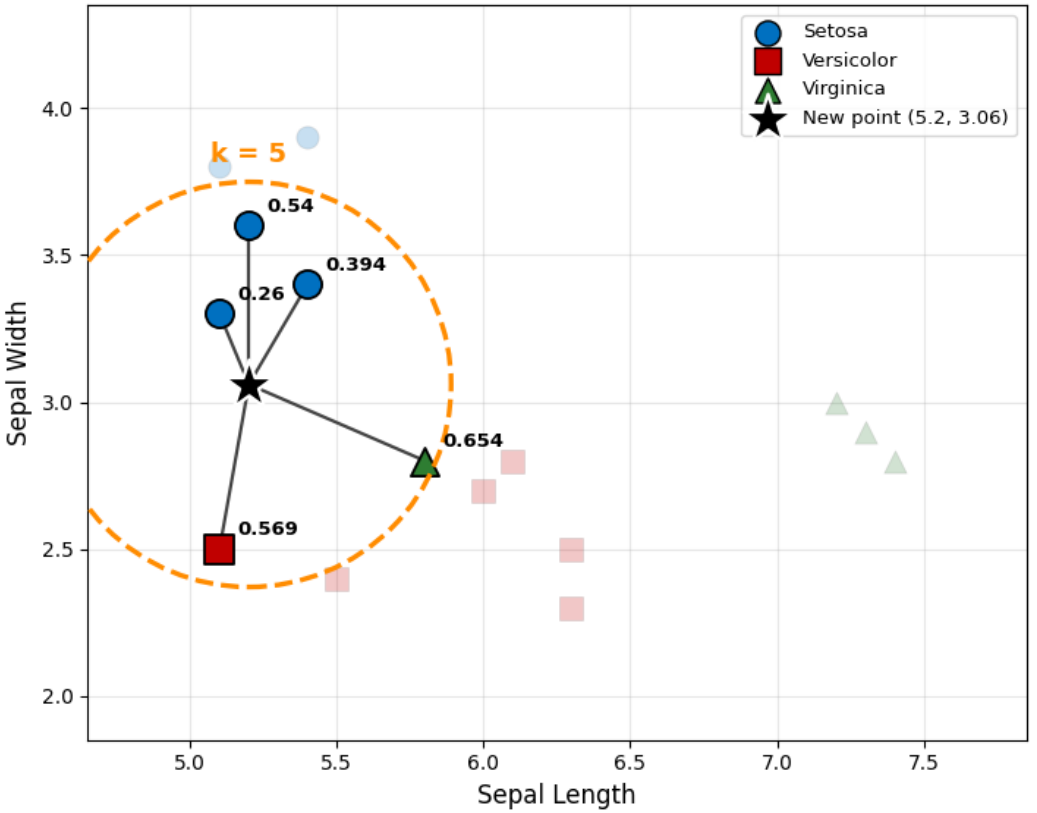
The Working of k-NN

Sepal Length	Sepal Width	Species	Distance	Rank
5.2	3.6	Setosa	0.540	3
5.1	3.8	Setosa	0.747	7
7.2	3.0	Virginica	2.001	13
5.4	3.4	Setosa	0.394	2
5.1	3.3	Setosa	0.260	1
5.4	3.9	Setosa	0.863	8
7.4	2.8	Virginica	2.215	15
6.1	2.8	Versicolor	0.937	10
7.3	2.9	Virginica	2.106	14
6.0	2.7	Versicolor	0.877	9
5.8	2.8	Virginica	0.654	5
6.3	2.3	Versicolor	1.337	12
5.1	2.5	Versicolor	0.569	4
6.3	2.5	Versicolor	1.234	11
5.5	2.4	Versicolor	0.725	6
5.2	3.06	**???**		

Setosa: 03
Virginica: 01
Versicolor: 01

Step 2: Rank the distances

K=5



Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species	Distance	Rank
5.2	3.6	Setosa	0.540	3
5.1	3.8	Setosa	0.747	7
7.2	3.0	Virginica	2.001	13
5.4	3.4	Setosa	0.394	2
5.1	3.3	Setosa	0.260	1
5.4	3.9	Setosa	0.863	8
7.4	2.8	Virginica	2.215	15
6.1	2.8	Versicolor	0.937	10
7.3	2.9	Virginica	2.106	14
6.0	2.7	Versicolor	0.877	9
5.8	2.8	Virginica	0.654	5
6.3	2.3	Versicolor	1.337	12
5.1	2.5	Versicolor	0.569	4
6.3	2.5	Versicolor	1.234	11
5.5	2.4	Versicolor	0.725	6
5.2	3.06	**Setosa**		

Setosa: 03
Virginica: 01
Versicolor: 01

↓

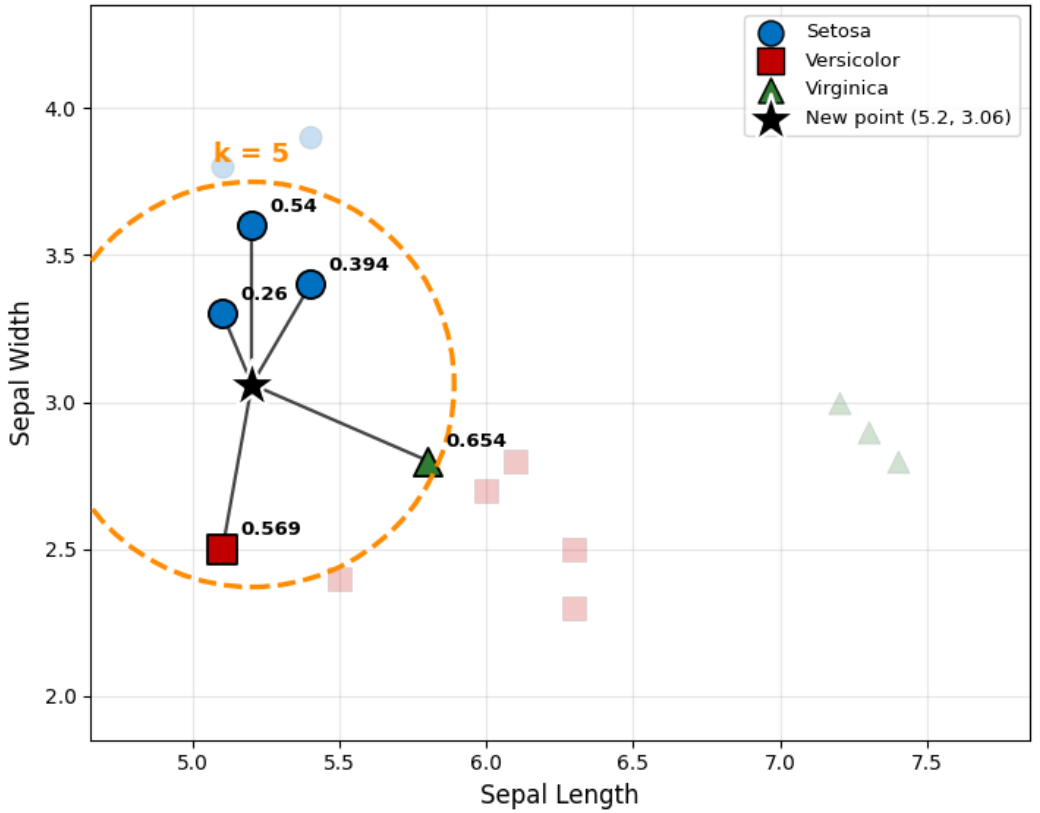
Max

↓

Setosa

Step 2: Rank the distances

K=5



Classification: k-Nearest Neighbors (k-NN)

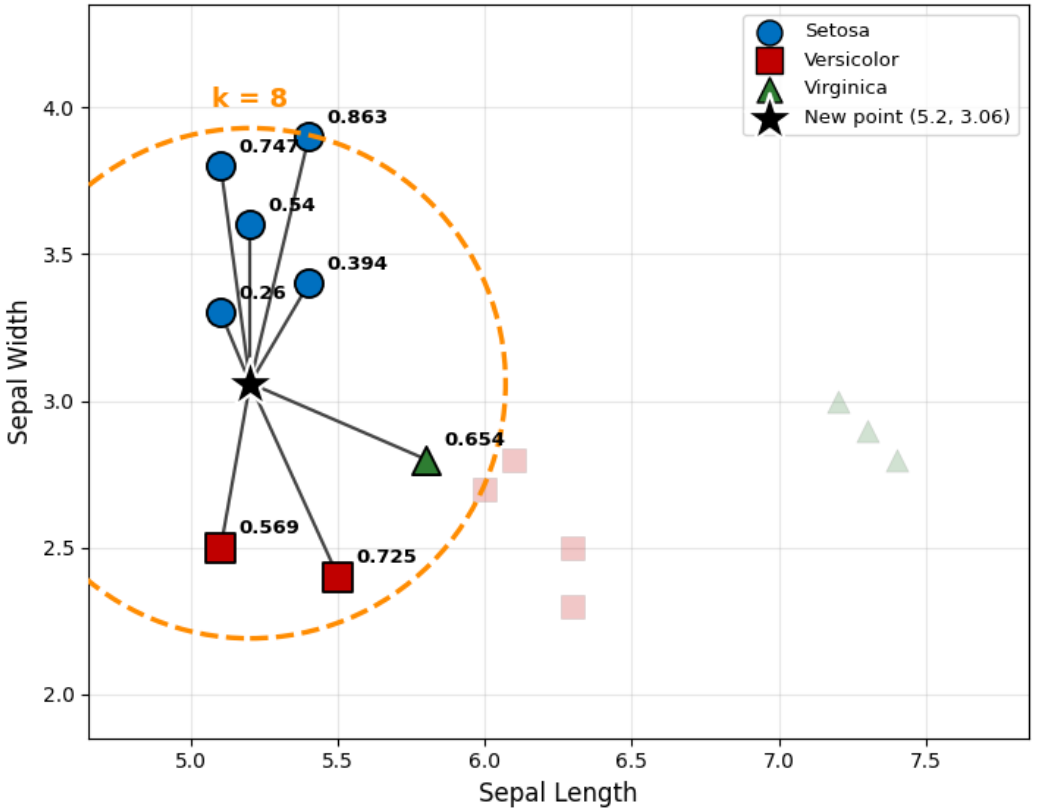
The Working of k-NN

Sepal Length	Sepal Width	Species	Distance	Rank
5.2	3.6	Setosa	0.540	3
5.1	3.8	Setosa	0.747	7
7.2	3.0	Virginica	2.001	13
5.4	3.4	Setosa	0.394	2
5.1	3.3	Setosa	0.260	1
5.4	3.9	Setosa	0.863	8
7.4	2.8	Virginica	2.215	15
6.1	2.8	Versicolor	0.937	10
7.3	2.9	Virginica	2.106	14
6.0	2.7	Versicolor	0.877	9
5.8	2.8	Virginica	0.654	5
6.3	2.3	Versicolor	1.337	12
5.1	2.5	Versicolor	0.569	4
6.3	2.5	Versicolor	1.234	11
5.5	2.4	Versicolor	0.725	6
5.2	3.06	**???**		

Setosa: 05
Virginica: 01
Versicolor: 02

Step 2: Rank the distances

K=8



Classification: k-Nearest Neighbors (k-NN)

The Working of k-NN

Sepal Length	Sepal Width	Species	Distance	Rank
5.2	3.6	Setosa	0.540	3
5.1	3.8	Setosa	0.747	7
7.2	3.0	Virginica	2.001	13
5.4	3.4	Setosa	0.394	2
5.1	3.3	Setosa	0.260	1
5.4	3.9	Setosa	0.863	8
7.4	2.8	Virginica	2.215	15
6.1	2.8	Versicolor	0.937	10
7.3	2.9	Virginica	2.106	14
6.0	2.7	Versicolor	0.877	9
5.8	2.8	Virginica	0.654	5
6.3	2.3	Versicolor	1.337	12
5.1	2.5	Versicolor	0.569	4
6.3	2.5	Versicolor	1.234	11
5.5	2.4	Versicolor	0.725	6
5.2	3.06	**Setosa**		

Setosa: 05
Virginica: 01
Versicolor: 02

↓

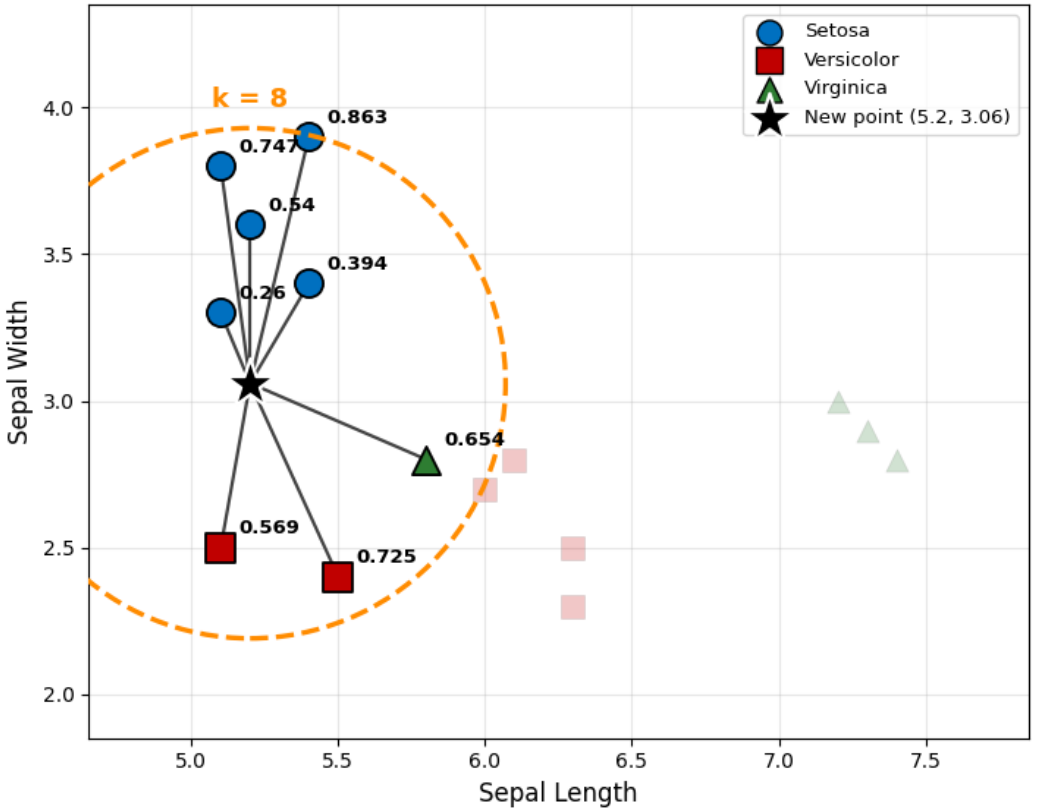
Max

↓

Setosa

Step 2: Rank the distances

K=8



Classification: k-Nearest Neighbors (k-NN)

Our Question For Today?

The scenario
A property assessor must place every house in a price band before the valuation is finalized.
They have the full record: size, quality, year built, garage, everything.
They do not yet have the price.

The question?

Given everything else about the house, is it **above** or **below** the market median?

How k-NN asks it differently?

- A tree asked: which yes/no questions separate the two groups?
- Naive Bayes asks: which price band was more likely to produce a house like this?
- k-NN asked: **which houses are most similar to this one?**